The Structure of Frames in Metric Linear Spaces and their Applications in Non-locally Convex Spaces

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Ethical Declaration

I certify that this research work is original, free from plagiarism, and in compliance with ethical standards.

Signature of Student: _____

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1 Aim of the Research Work

Frame theory offers stable and redundant representations that extend the classical notion of bases and orthonormal systems. Frames play a significant role in non-harmonic Fourier analysis and wavelet theory, with extensive applications in data representation, signal processing, and harmonic analysis. Traditional frame theory has been largely developed within the framework of Hilbert spaces, where inner product structures and dual spaces are essential. As a generalization of bases in Hilbert and Banach spaces, frame theory has attracted increasing attention since the 1980s, particularly following the development of wavelet methods and their applications in computer science. Extending frame theory to more abstract ambient spaces, such as metric linear spaces, offers the potential to connect classical frame concepts with topological vector spaces and quasi-Banach spaces.

Generalizing normed spaces broadly is exactly what metric linear spaces do. In the twentieth century, early studies mostly just looked at cases that were locally convex. Developing Banach space theory and modern functional analysis is what this eventually led to. Sometimes non-locally convex spaces just pop up naturally. You see them in probability theory and stochastic processes. Approximation theory, the theory of analytic functions, and integral operators have them too. Standard techniques from Hilbert and Banach space theory end up being pretty useless or impossible to apply in these specific settings. This inadequacy happens mostly because things like local convexity and norms are totally missing. Guaranteed non-trivial dual spaces are also often absent.

Sitting in a highly important spot within functional analysis is the theory of metric linear spaces. It acts as a very natural way to generalize locally convex and normed spaces. Discussing the subject's early developments is how this dissertation starts. Highlighting what Banach and Fréchet contributed foundationally is done first. Moving toward frameworks where local convexity is not relied upon at all happens right after that. Rolewicz did some very comprehensive work. Building heavily upon that work, developing the structural theory of metric linear spaces systematically is what the study does. Emphasizing F^*-spaces and F-spaces in particular is a big part of it.

Undertaking a detailed analysis of generalized convexity concepts inside metric linear spaces happens next. We examine p-convexity and pseudoconvexity notions. Classical results from Day and Aoki and Rolewicz get looked at as well. Understanding the geometry of these spaces heavily relies on those

exact results. There are intrinsic structural difficulties that the study also has to address. Unconditional convergence being completely absent is one of them. Dual spaces potentially being trivial is another huge issue. Extending Hilbert space techniques directly is complicated greatly by these exact factors.

The dissertation subsequently turns to the theory of frames in abstract settings. Beginning with their emergence in nonharmonic Fourier analysis and wavelet theory, the evolution of frame concepts beyond Hilbert spaces is explored. Various approaches to defining frames in Banach spaces, p-Banach spaces, and general metric spaces are reviewed and compared.

A huge portion of this thesis is focused on discussion frames in Fréchet spaces. We spend a considerable amount of time systematically reviewing the contemporary developments in frame theory and its abstractions. We first discuss Fréchet spaces as their own entity and differentiate them from F-spaces that were expounded upon earlier. We show how historically the study of these ambient spaces emerged and how both of them are metric linear spaces. We study the various equivalent formations of a Fréchet space, hoping to see why one might be better than others for a rich theory of frames in the space.

The sections delineating the theory of Fréchet frames are based primarily on the body of work in the area by Pilipović and Stoeva, we talk about projective limits, how every complete locally convex space is equivalent to the projective limit of a sequence of Banach spaces. We use this property of Fréchet spaces to define analogous versions of Banach sequence spaces and Frames in Fréchet spaces.

Finally we spend some time on generalization of Fréchet frames and their applcation to specific ambient spaces like Schwatrz spaces and Gelfand-Shilov spaces. We also discuss some recent results in the theory of frames in non-locally convex p-Banach spaces.

Through this survey, common themes and methodological strategies are identified, shedding light on both the challenges and prospects for developing meaningful frame structures in non-locally convex metric linear spaces.

2 Review of Literature

Metric Linear Spaces

The notion of a metric linear space traces its origins to the work of Fréchet in 1926 [39]. Subsequent progress in the coming decades, particularly by Banach and his collaborators, significantly shaped the foundations of linear metric spaces [58, 65, 72, 73, 82, 83, 84]. A landmark contribution is Banach's monograph *Théorie des opérations linéaires* (1932), where the theory was developed mainly under assumptions of local convexity, laying the groundwork for modern functional analysis [10, 9, 45, 31].

Let X be a set. A mapping $\rho : X \times X \to \mathbb{R}$ is called a *metric* if, for all $r, s, t \in X$, the following conditions hold:

(m1) $\rho(r, s) \geq 0$, and $\rho(r, s) = 0$ if and only if $r = s$;

(m2) $\rho(r, s) = \rho(s, r)$;

(m3) $\rho(r, t) \leq \rho(r, s) + \rho(s, t)$.

Throughout this thesis, all vector spaces are assumed to be real or complex. A vector space X equipped with a metric ρ is called a *metric linear space* if the operations of vector addition and scalar multiplication are continuous with respect to the topology induced by ρ.

A comprehensive account of this subject can be found in Rolewicz's monograph *Metric Linear Spaces* [85], as well as the lecture notes by Kalton and collaborators [53, 52].

Given two metrics ρ and ρ', we say that ρ is *stronger* than ρ' if convergence in ρ implies convergence in ρ'. If this implication holds in both directions, the metrics are said to be *equivalent*.

A metric ρ is called *translation invariant* if

$$\rho(r + t, s + t) = \rho(r, s), \qquad \forall r, s, t \in X.$$

A subset $W \subseteq X$ is said to be *balanced* whenever $\alpha W \subseteq U$ for all scalars α satisfying $|\alpha| \leq 1$.

Banach also introduced the notion of an F^*-space, which plays a central role in the theory of metric linear spaces.

Let X be a vector space. A function $\| \cdot \| : X \to \mathbb{R}$ is called an *F-norm* if it satisfies:

(N1) $\|r\| = 0$ if and only if $r = 0$;

(N2) $\|\alpha\, r\| = \|r\|$ for all scalars α with $|\alpha| = 1$;

(N3) $\|r + s\| \leq \|r\| + \|s\|$;

(N4) $\|\alpha_n\, r\| \to 0$ whenever $\alpha_n \to 0$;

(N5) $\|\alpha\, r_n\| \to 0$ whenever $r_n \to 0$;

(N6) $\|\alpha_n\, r_n\| \to 0$ whenever $\alpha_n \to 0$ and $r_n \to 0$.

If condition (N1) is omitted, $\|\cdot\|$ is referred to as a *pseudonorm*.

Proposition 1. *If* $\|r_n - r\| \to 0$ *and* $\alpha_n \to \alpha$, *then* $\|\alpha_n\, r_n - \alpha\, x\| \to 0$.

Each F-norm induces a translation invariant metric defined by

$$\rho(r, s) = \|r - s\|.$$

Conversely, every translation invariant metric on a real metric linear space gives rise to an F-norm via $\|r\| = \rho(r, 0)$.

A vector space equipped with an F-norm is called an *F^*-space*. If it is complete with respect to the induced metric, it is called an *F-space*.

Metric linear spaces constitute a subclass of metrizable topological vector spaces. A *topological linear space* is a vector space endowed with a Hausdorff topology in which both vector addition and scalar multiplication are continuous.

In 1936, Kakutani and Birkhoff independently characterized metrizable topological vector spaces [51, 13]. Their result asserts that a topological vector space is metrizable if and only if it admits a countable neighborhood basis at the origin. An equivalent formulation emphasizes the existence of an invariant metric [85, 57].

Theorem 2 (Kakutani, 1936 [51])**.** *Let* (X, ρ) *be a metric linear space. Then there exists a translation invariant metric* ρ' *equivalent to* ρ.

As a consequence, the categories of metric linear spaces and F^*-spaces coincide.

Modulars

Important examples of metric linear spaces, use the concept of a modular, developed in the works of Nakano, Musielak, and Orlicz [71, 69, 70, 68].

Definition 3 (Modular)**.** Let X be a linear space. A function $\rho : X \to [0, \infty]$ is called a *modular* if it satisfies:

(MD1) $\rho(r) = 0$ if and only if $r = 0$;

(MD2) $\rho(\alpha\, r) = \rho(r)$ whenever $|\alpha| = 1$;

(MD3) $\rho(\alpha\, r + \beta\, s) \leq \rho(r) + \rho(s)$ for $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$;

(MD4) $\rho(\alpha_n\, r) \to 0$ whenever $\alpha_n \to 0$ and $\rho(r) < \infty$.

If additionally

(MD5) $\rho(\alpha\, r_n) \to 0$ whenever $\rho(r_n) \to 0$,

then ρ is called a *metrizing modular*.

In what follows, the term *modular* always refers to a metrizing modular. Let

$$X_\rho = \{r \in X : \rho(k\, r) < \infty \text{ for some } k > 0\}.$$

Theorem 4 (Musielak–Orlicz, 1959 [69])**.** *If X is equipped with a metrizing modular ρ, then there exists an F-norm $\|\cdot\|$ on X_ρ such that $\|r_n\| \to 0$ if and only if $\rho(r_n) \to 0$.*

The theory of modulars naturally leads to Orlicz spaces, which provide important examples of metric linear spaces and play a key role in the study of non-locally convex analysis.

Convexity

Convexity has long been recognized as a fundamental structural concept in analysis. Minkowski's pioneering work at the end of the nineteenth century introduced convex sets in Euclidean spaces and examined their geometric properties [67]. The systematic use of convexity in functional analysis emerged later, notably in the Krein–Milman theorem [62]. A classical reference is Valentine's monograph [96].

Let X be a metric linear space. A set $Z \subset X$ is called *starlike* if $tZ \subset Z$ for all $0 < t \leq 1$.

Rolewicz introduced the *modulus of concavity* of a starlike set Z by

$$c(Z) = \inf\{s > 0 : Z + Z \subset sZ\}.$$

If $c(Z) < \infty$, the set Z is called *pseudoconvex*. Every convex set is pseudoconvex with $c(Z) = 2$, although the converse need not hold.

A metric linear space is said to be *locally pseudoconvex* if it admits a neighborhood basis at the origin consisting of pseudoconvex sets. If the moduli of concavity are uniformly bounded by $2^{1/p}$, the space is called *locally p-convex* [4, 63].

The foundational works of Turpin, Simmons, and Łelazko [98, 94, 89] established the following characterization.

Theorem 5. *Let X be locally pseudoconvex. Then its topology can be generated by a sequence of p_n-homogeneous F-pseudonorms. If X is locally p-convex, one may take $p_n = p$ for all n.*

These results highlight the close connection between convexity-type conditions and the existence of suitable quasi-norm structures in metric linear spaces. The interested reader may refer to the following works discussing pseudoconvexity [6, 95, 88, 49, 34].

Frames

Frames were introduced as a flexible alternative to bases, allowing redundant yet stable representations of elements in a space. Unlike Schauder bases, frame expansions are generally non-unique.

The concept was first formalized by Duffin and Schaeffer in their study of non-harmonic Fourier series [33, 32]. A systematic exposition was later provided by Young [97]. Subsequent developments, particularly in wavelet theory, demonstrated the importance of frames in signal processing and harmonic analysis [30, 29, 27, 28]. A detailed study can be found in the classic monolith on the subject by Christensen [22].

Definition 6 (Frame)**.** Let $\mathfrak{H}$ be a Hilbert space. A collection $\{g_j\}_{j\in J} \subset \mathfrak{H}$ is said to be a *frame* for $\mathfrak{H}$ if there exist positive constants $\mathcal{A}$ and $\mathcal{B}$ such that

$$\mathcal{A}\|g\|^2 \leq \sum_{j\in J} \big|\langle g, g_j\rangle\big|^2 \leq \mathcal{B}\|g\|^2, \qquad \text{for all } g \in \mathfrak{H}.$$

Frames have since been generalized in several directions, notably to Banach spaces and beyond, motivating the study of atomic decompositions, p-frames, and frame-like structures in non-locally convex settings.

3 Objectives

Pushing frame theory completely outside of where it usually sits comfortably is the main purpose of this work. Classical theory has roots deep inside Hilbert spaces. However, seeing how far it might get carried into settings that are much more general is the intention here. These settings are particularly metric linear spaces. Spaces like Banach and quasi-Banach are included, and even spaces lacking local convexity are looked at. In these areas, analytical toolkits that people are familiar with do not always do what is expected. Sometimes, they just fail completely.

Understanding the structure underneath these spaces takes up a substantial part of the study. The discussion primarily focuses on F-spaces and F^*-spaces. It begins with early ideas from Fréchet and Banach. Then, it moves right through Rolewicz's developments which are more systematic. Notions like p-convexity and pseudoconvexity are introduced along the way. But they are also examined for what it is that they actually do. Namely, recovering a sense of geometric control in environments where having convexity is no longer something that is guaranteed is how they help.

Confronting certain difficulties entirely head-on becomes necessary at the same time. Dual spaces have the ability to collapse into something trivial. Unconditional convergence might fail in many of these spaces. Arguments from standard Hilbert space simply stop working. How frame theory has to adapt when these limitations are present is one of the guiding questions, then. Also, figuring out exactly which parts of the classical theory can be realistically salvaged is another.

Tracing how frames evolved originally requires the work to take a step back. This goes from wavelet theory straight from non-harmonic Fourier series. Following their gradual extension into p-Banach and Banach spaces happens next. Quite deliberately, the focus shifts toward Fréchet spaces from there. Treating these as just another generalization is not done. Instead, they are seen as a setting possessing its own internal logic. It sits somewhere exactly between flexibility and structure. A key thread forming in the discussion is understanding why they are frequently suited better for frame constructions, as well as how general F-spaces relate to Fréchet spaces.

Attempting to build appropriate sequence spaces in the Fréchet setting is what the study then does by using the perspective of projective limits. Frame definitions that correspond are also built. The work of Pilipović and Stoeva is drawn upon heavily for this. Because many properties that are familiar have

to be reinterpreted instead of just assumed, this process operates less like a straightforward extension. It operates far more like a careful reconstruction.

Widening the scope again toward the end happens to include generalizing Fréchet frames further. Their behavior inside concrete function spaces is looked at. Gelfand-Shilov and Schwartz spaces are examples of this. Bringing recent developments regarding frame theory for non-locally convex p-Banach spaces into the picture is also done. Illustrating just how active and completely unsettled this specific area still manages to be is partly the reason for doing so.

Looking for patterns rather than just collecting results is the aim in a broader sense. We want to know what breaks down. We want to know what carries over. Figuring out what new ideas are required to bridge the gap is necessary. Leading to a significantly clearer sense of how extending frame theory meaningfully is possible is the hope here. A better understanding of where the actual opportunities and real obstacles lie is expected along with it.

4 Methodology

Statement of the Problem

The main reason we are doing this research comes from how there isn't a really good, organized way to describe the structure of frames in general metric linear spaces. Most of the time, frame theory grew up inside Hilbert spaces. In those spaces, things like inner products and self-duality are super important. Sure, people have suggested ways to move this to Banach spaces with things like atomic decompositions and p-frames. But these ways usually need the dual spaces to be pretty rich to work at all.

On the flip side, lots of metric linear spaces just do not have these nice structural properties. This is especially true for non-locally convex spaces and F-spaces. They might not even have norms, and their dual spaces are often completely trivial or just not enough. Because of this, classical tools like the Hahn-Banach theorem simply cannot be applied, and unconditional convergence is out of the question. Therefore, when the geometric framework of Hilbert and Banach spaces disappears, it remains quite unclear exactly which basic frame properties actually survive. Addressing this theoretical gap by systematically looking at frame-like structures in non-locally convex and metric linear settings is what this dissertation tries to do.

Scope of the Research

We place this study inside the wider area of functional analysis. The specific focus happens to be on how **metric linear spaces** interact with **frame theory**.

- Ambient spaces like F-spaces, Orlicz spaces, and metric linear spaces are what the study looks into primarily. Spaces that have non-locally convex topologies are also included, like L^p spaces for $0 < p < 1$.
- Moving past the usual Hilbert space setup, it explores generalized ideas like metric frames and p-frames. At the same time, it purposely only pays attention to topological and analytical sides, leaving numerical or computational implementations aside.
- By proposing conceptual extensions of frame theory that can apply to very abstract non-locally convex and metric linear spaces, the research

wants to push forward what is currently happening in locally convex and quasi-Banach frameworks.

Aims and Objectives

Making a frame-theoretic framework that rests fundamentally on topological and metric structures instead of inner product geometry is the central aim of this dissertation. The specific objectives that we have are the following:

1. **To examine structural characteristics** belonging to frames inside spaces that lack inner product structures or local convexity.
2. **To determine minimal assumptions** where getting stability and reconstruction in general metric linear spaces becomes possible.
3. **To extend and unify** concepts already out there, like Banach frames and atomic decompositions, bringing them into a metric-linear framework.
4. **To analyze the limitations** that happen because unconditional convergence and non-trivial dual spaces are missing, as well as checking if frame-like representations are even feasible in settings like these.

Research Questions

Trying to reach those objectives mentioned above, the following research questions are addressed by this study:

1. When neither norms nor inner products are available in metric linear spaces, how can the idea of a frame be reformatted in a way that makes sense?
2. In spaces where duals are trivial, like L^p spaces for $0 < p < 1$, how can we define frame-like structures, and what is the role the dual space plays in reconstruction?
3. How might we deal with issues that have to do with the unconditional convergence of series expansions inside metric linear spaces?

4. Classical properties of frames, like having frame operators exist, persist to what extent when put into non-locally convex and metric linear environments?

5. Is it possible to take existing frame generalizations from locally convex, quasi-Banach, and Banach spaces and extend them further into topological vector spaces and general metric linear spaces?

Hypothesis

This research does not push one definitive claim. Instead, it operates under the working hypothesis that **inner product structures might not be intrinsically tied to the essential reconstruction and stability properties of frames, which might instead rely on the topological and metric features underlying the space**.

Specifically, whether a theory of frames in metric linear spaces that somewhat parallels the classic Hilbert space framework can be supported by using invariant metrics and F-norms is what the study explores.

Research Methodology

Because of the rules for **Doctrinal Research (Type C)**, a purely analytical and theoretical methodology is adopted by this study. It focuses hard on extending and critically examining mathematical theories that are already established.

Theoretical Issues Addressed

Reconciling how frame theory is usually constructive (relying on duality and inner products) with the non-classical features inherent to metric linear spaces is the main theoretical challenge. These features include having non-trivial duals and lacking local convexity. To what extent definitions and properties of frames can actually be extended to such abstract places in a meaningful way is evaluated by this research.

Primary and Secondary Sources

Looking systematically through authoritative mathematical resources is how the literature review got done:

- MathSciNet was used to pinpoint primary references.
- Major academic publishers like the American Mathematical Society, Elsevier, and Springer, along with Google Scholar, provided supplementary sources.
- The following thematic areas were the focus of the review:
  - Frames in Banach and Hilbert spaces
  - Banach frames and atomic decompositions
  - Structures related to frames in non-locally convex and metric linear spaces
  - Topological vector spaces and metric linear spaces

Method of Study

Both **comparative** and **analytical** describes the adopted methodology:

1. **Comparative Analysis:** To find out which properties do not change when the geometric structure shifts, we compare how frames are defined across different ambient spaces.
2. **Axiomatic Development:** We look at existing ways frame theory has been built axiomatically. The goal here is making analogous definitions that fit metric linear spaces well.
3. **Counterexample Analysis:** To see how robust the proposed frame constructions are and find their limits, classical pathological examples are used, such as L^p spaces for $0 < p < 1$.

Limitations

1. **Structural Limitations:** Building dual frames in the classical sense gets restricted in certain metric linear spaces because continuous linear functionals that are non-trivial are completely absent.

2. **Convergence Issues:** Guaranteeing that frame expansions will converge is a big challenge due to the lack of unconditional convergence in general F-spaces.

Results and Discussions

5 Frames in Hilbert Spaces

Introduction

Hilbert spaces constitute one of the most important mathematical frameworks in functional analysis and its applications. They provide a natural setting for studying infinite-dimensional vector spaces endowed with an inner product, which allows for the notions of length, angle, and orthogonality. Classical examples include the Euclidean space $\mathbb{R}^n$, the sequence space ℓ^2, and function spaces such as $L^2(\Omega)$.

A central problem in Hilbert space theory is the representation of vectors in terms of simpler or more structured elements. Orthonormal bases play a fundamental role in this regard, as they allow every element of the space to be written uniquely as a convergent series of scalar multiples of basis vectors. However, in many practical and theoretical situations, the strict requirements imposed by orthonormal bases are too restrictive.

Frames were introduced to overcome these limitations by allowing redundant, yet stable, representations of vectors in Hilbert spaces. Unlike bases, frames do not require linear independence or uniqueness of expansion. This redundancy turns out to be advantageous in numerous applications, including signal processing, image analysis, sampling theory, wavelet analysis, and quantum mechanics.

This chapter presents a basic and self-contained introduction to frames in Hilbert spaces. We begin by recalling the concept of bases and motivate the need for frames. We then introduce the definition of frames, discuss their basic properties, and examine several illustrative examples. The chapter concludes with a discussion of frame operators and reconstruction formulas.

Hilbert Spaces and Orthonormal Bases

Let $\mathfrak{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and the induced norm $\|g\| = \sqrt{\langle g, g \rangle}$. A sequence $\{e_k\}_{k=1}^{\infty} \subset \mathfrak{H}$ is called an *orthonormal sequence* if

$$\langle e_k, e_l \rangle = \begin{cases} 1, & k = l, \\ 0, & k \neq l. \end{cases}$$

An orthonormal sequence is called an *orthonormal basis* if its closed linear span equals the entire space $\mathfrak{H}$. In this case, every element $g \in \mathfrak{H}$ admits the expansion

$$g = \sum_{k=1}^{\infty} \langle g, e_k \rangle e_k,$$

where the series converges in the norm of $\mathfrak{H}$. Moreover, the Parseval identity holds:

$$\|g\|^2 = \sum_{k=1}^{\infty} |\langle g, e_k \rangle|^2.$$

Orthonormal bases provide elegant and efficient representations. However, they are extremely sensitive to perturbations. Removing or modifying even a single basis element may destroy completeness and hence the ability to represent all vectors in the space.

Motivation for Frames

In numerous applications, especially in signal and image processing, robustness and stability are of primary importance. Real-world data is often corrupted by noise, transmission errors, or partial loss of information. In such contexts, representations based on orthonormal bases may fail to be reliable.

Frames address this issue by allowing redundancy. The idea is to use more vectors than strictly necessary to represent elements of the space, while maintaining control over the size of the coefficients. This redundancy ensures that the loss of some frame coefficients does not necessarily prevent accurate reconstruction.

From a mathematical perspective, frames generalize orthonormal bases by replacing the equality in Parseval's identity with two-sided inequalities. This relaxation leads to a flexible and powerful representation theory.

Definition of Frames

Definition 7. Let $\mathfrak{H}$ be a Hilbert space. A sequence $\{g_k\}_{k=1}^{\infty} \subset \mathfrak{H}$ is called a *frame* for $\mathfrak{H}$ if there exist constants $\mathcal{A}, \mathcal{B} > 0$ such that

$$\mathcal{A}\|g\|^2 \leq \sum_{k=1}^{\infty} |\langle g, g_k \rangle|^2 \leq \mathcal{B}\|g\|^2, \quad \forall g \in \mathfrak{H}. \tag{1}$$

The constants $\mathcal{A}$ and $\mathcal{B}$ are known as the *frame bounds*. The largest possible lower bound and the smallest possible upper bound are referred to as the optimal frame bounds.

The frame inequality ensures that the frame coefficients $\{\langle g, g_k\rangle\}$ provide a stable description of the vector g. The lower bound guarantees that no nonzero vector can have all coefficients close to zero, while the upper bound prevents the coefficients from growing uncontrollably.

Special Classes of Frames

5.1 Tight Frames

A frame is said to be a *tight frame* when its lower and upper frame bounds coincide, that is, $\mathcal{A} = \mathcal{B}$. In this case, the frame inequality simplifies to

$$\sum_{k=1}^{\infty} |\langle g, g_k\rangle|^2 = \mathcal{A}\|g\|^2, \quad \forall g \in \mathfrak{H}.$$

Tight frames resemble orthonormal bases in many respects, particularly in their reconstruction formulas. When $A = 1$, the frame is called a *Parseval frame*.

5.2 Exact Frames

A frame is said to be *exact* if the removal of any single element destroys the frame property. Exact frames may be viewed as the closest analogues of bases within the class of frames.

Basic Properties of Frames

Every frame $\{g_k\}$ for $\mathfrak{H}$ is necessarily complete, meaning that

$$\overline{\text{span}}\{g_k : k \in \mathbb{N}\} = \mathfrak{H}.$$

However, frames need not be linearly independent. This lack of independence is precisely what allows redundancy.

Another important observation is that every orthonormal basis is a frame with bounds $\mathcal{A} = \mathcal{B} = 1$. Hence, frames truly generalize the classical notion of bases.

Examples of Frames

Example 8. Let $\{e_k\}_{k=1}^{\infty}$ be an orthonormal basis for $\mathfrak{H}$.

1. Repeating each basis element twice yields the sequence
$$\{e_1, e_1, e_2, e_2, \ldots\},$$
which constitutes a tight frame with frame bound $\mathcal{A} = 2$.

2. Scaling the basis elements as $\{\alpha_k e_k\}$, where $0 < m \leq |\alpha_k| \leq M < \infty$, yields a frame with bounds $\mathcal{A} = m^2$ and $\mathcal{B} = M^2$.

3. Any finite orthonormal set is not a frame for $\mathfrak{H}$, but it forms a frame for its closed linear span.

These examples demonstrate how frames arise naturally from modifications of orthonormal bases.

Analysis and Synthesis Operators

Given a frame $\{g_k\}$ for $\mathfrak{H}$, the mapping
$$T^* : \mathfrak{H} \to \ell^2, \quad T^* f = \{\langle g, g_k \rangle\}_{k=1}^{\infty},$$
is called the *analysis operator*. The adjoint operator
$$T : \ell^2 \to \mathfrak{H}, \quad T(\{c_k\}) = \sum_{k=1}^{\infty} c_k g_k,$$
is known as the *synthesis operator*.

Both operators are bounded and linear, and they play a central role in frame theory.

The Frame Operator and Reconstruction

The composition $S = TT^*$ defines the *frame operator*, given by
$$Sg = \sum_{k=1}^{\infty} \langle g, g_k \rangle g_k.$$

The frame operator is bounded, positive, self-adjoint, and invertible. These properties ensure that every vector in $\mathfrak{H}$ can be reconstructed from its frame coefficients.

Theorem 9 (Frame Reconstruction Formula)**.** *Let* $\{g_k\}$ *be a frame for* $\mathfrak{H}$ *with frame operator* S*. Then every* $g \in \mathfrak{H}$ *admits the representation*

$$g = \sum_{k=1}^{\infty} \langle g, S^{-1} g_k \rangle g_k = \sum_{k=1}^{\infty} \langle g, g_k \rangle S^{-1} g_k,$$

where both series converge in norm.

The sequence $\{S^{-1} g_k\}$ is itself a frame, called the *canonical dual frame.*

Concluding Remarks

Frames in Hilbert spaces provide a powerful and flexible framework for vector representations. By allowing redundancy while preserving stability, frames bridge the gap between theoretical elegance and practical robustness. Their rich structure and wide applicability make them an indispensable tool in modern analysis and applied mathematics.

Why Frames?

Frames provide a robust alternative to bases. Unlike bases, they are stable under bounded surjective operators. In contrast, a basis remains a basis under a linear operator only if the operator is bounded and invertible.

In applications, signal representations often require resilience to noise and local perturbations. Frames naturally offer redundancy, which enhances stability and numerical robustness.

Daubechies, Grossmann, and Meyer [30] emphasized that strict orthogonality may introduce undesirable artefacts. For instance, orthonormal expansions may exhibit global sensitivity to localized signal modifications or artificial discontinuities due to domain partitioning.

Frames mitigate these issues by allowing flexible, overlapping representations that preserve locality while maintaining stable reconstruction.

Generalizations

Let X be a Banach space with dual X^*. In many applications, one seeks frame- like expansions beyond Hilbert spaces. To this end, Banach sequence spaces $X_d \subset \mathbb{C}^{\mathbb{N}}$ play a central role.

Atomic Decompositions

Atomic decompositions extend frame expansions to Banach spaces [25, 35, 23, 41]. More recently, Kaushik and Sharma have carried out extensive investigations into atomic decompositions in a series of joint works [54, 55, 56].

Definition 10. A pair $(\{h_k\}, \{g_k\}) \subset X^* \times X$ is called an *atomic decomposition* of X with respect to X_d if:

1. $\{h_k(g)\} \in X_d$ for all $g \in X$,
2. There exist $A, B > 0$ such that

$$\mathcal{A}\|g\| \leq \|\{h_k(g)\}\|_{X_d} \leq \mathcal{B}\|g\|,$$

3. Every $g \in X$ admits the expansion $g = \sum_{k=1}^{\infty} h_k(g) g_k$.

Banach Frames

Initial studies into Banach frame generalizations of frames are by Christensen et. al. [20, 26].

Definition 11. A pair $(\{h_k\}, S)$ is called a *Banach frame* for X with respect to X_d if $S : X_d \to X$ is bounded and provides stable reconstruction:

$$S(\{h_k(g)\}) = g, \quad \forall g \in X.$$

Banach frames allow reconstruction through bounded operators rather than explicit series, making them more flexible in non-Hilbert settings.

5.3 p-Frames

The work of Aldroubi et. al. introduces the following idea for a frame [5, 24].

Let $p \in (1, \infty)$. A sequence $\{h_k\} \subset X^*$ is called a *p-frame* if

$$\mathcal{A}\|g\| \leq \left(\sum_{k=1}^{\infty} |h_k(g)|^p \right)^{1/p} \leq \mathcal{B}\|g\|.$$

Such frames yield expansions in shift-invariant subspaces of $L^p(\mathbb{R})$ and retain stability across all $p \in (1, \infty)$.

6 Metric Linear Spaces

Metric Linear Spaces and Invariant Metrics

A *metric linear space* (X, ρ) is a linear space equipped with a metric ρ such that both vector addition and scalar multiplication are continuous operations.

A metric ρ is called *translation invariant* if

$$\rho(r + t, s + t) = \rho(r, s), \quad \forall r, s, t \in X.$$

Theorem 12 (Kakutani)**.** *Every metric linear space admits an equivalent invariant metric.*

6.1 F-Norms and F-Spaces

An *F-norm* is a function $\|\cdot\|$ satisfying generalized norm conditions compatible with metric linear structures.

Every invariant metric induces an F-norm, and every complete F-space is called an *F-space*.

Modular Spaces

A *modular* ρ is a functional satisfying convexity-type conditions. The associated space

$$X_\rho = \{r : \rho(k\, r) < \infty \text{ for some } k > 0\}$$

forms a linear subspace.

Theorem 13 (Musielak–Orlicz)**.** *Every metrizing modular induces an equivalent F-norm on X_ρ.*

7 Fréchet Spaces

Fréchet spaces are closely related to the 'metric linear spaces' that we have discussed before. We shall study the expansion of the idea of frames into the Fréchet spaces as an important step toward the generalisation of frames into the more abstract spaces from its original conception in the Hilbert spaces. Let us start with understanding some basic facts about Fréchet spaces.

7.1 Fréchet Spaces

The notion of a Fréchet space does not appear all at once in its modern form; rather, it grows out of the early attempts, around the turn of the twentieth century, to formalize ideas of convergence beyond the familiar Euclidean setting. A key figure here is Maurice Fréchet, whose 1906 thesis introduced what he called an *abstract space*, equipped with a generalized distance function. This was, in a sense, a decisive break from thinking only in terms of concrete subsets of $\mathbb{R}^n$, moving instead toward axiomatic descriptions.

From this starting point, Fréchet arrived at what we now call a *metric space*. It allowed one to discuss convergence, continuity, and compactness in settings that no longer resembled classical geometry in any trivial way. These ideas, somewhat unexpectedly at the time, became foundational for both topology and functional analysis.

Over the following decades, the picture was refined by several mathematicians, including Hausdorff, Riesz, and Banach. A lot of the work in this area was undertaken at the Lwów School of Mathematics (of which Banach was a co-founder), especially during the period between the two world wars. In particular:

- Banach introduced the concept of a *Banach space*, which is now familiar to most of us who have studied college mathematics, a normed vector space that is complete.
- At the same time, examples arising from analysis, especially spaces of functions, began to show that not every "natural" space admits a norm, even though it still behaves well from a topological standpoint, like Sobolev spaces.

It is in trying to deal with such examples that the idea of a *Fréchet space* gradually took shape. The origin of this terminology is somewhat interesting. While Fréchet worked with metric ideas, Banach clarified the role of

norms, and only later did the term "Fréchet space" come to denote a broader class of spaces—those that are complete and metrizable, but not necessarily normable. However, the first time someone used the term Fréchet spaces was Banach in response to Fréchet calling normed vector spaces, *Banach spaces* [10, 38].

In modern treatments, one usually builds local convexity into the definition. This reflects developments in the theory of locally convex spaces, particularly as organized in the work of Bourbaki, where such assumptions turn out to be both natural and technically useful [16].

Motivation

While Banach spaces form a very robust framework, they are, in practice, too rigid for certain important constructions. For example:

- The space $C^\infty(\Omega)$ of smooth functions on an open set $\Omega \subseteq \mathbb{R}^n$ does not admit a single norm that simultaneously controls all derivatives.
- Likewise, spaces of holomorphic functions, when given the topology of uniform convergence on compact subsets, fail to be normable.

Even so, these spaces are far from pathological. What actually happens is that their topology can be described using *countably many seminorms*, rather than a single norm. This observation leads more or less directly to the class of Fréchet spaces: they retain enough structure to support analysis, but are flexible enough to include these examples that fall outside the Banach framework.

7.2 Definition of Fréchet Spaces

We shall first give the definition of topological vector spaces which are one of the most abstract spaces one can do meaningful analysis on. We also will see how these ideas are further used in our definition of Fréchet frames.

Definition 14 (Topological vector space)**.** A *topological vector space* (TVS) X over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ is a vector space together with a topology for which the maps

$$X \times X \to X, \quad (x, y) \mapsto x + y,$$

and

$$\mathbb{K} \times X \to X, \quad (\lambda, x) \mapsto \lambda x,$$

are continuous.

Definition 15 (Local convexity)**.** One says that a TVS X is *locally convex* if there exists a neighborhood base at 0 consisting entirely of convex sets. (In practice, this condition ends up being quite restrictive, though in a useful way.)

An equivalent way to say this is a TVS, X is locally-convex if it is a Hausdorff space such that every neighborhood of $x \in X$ contains a convex neighborhood of x. You will notice our definition of local convexity is a TVS is the same as that in a metric linear space which we encountered in the earlier chapters.

A topology with a base of convex neighborhoods of 0 such that vector addition and scalar multiplication is continuous with this topology is called a *locally-convex topology.*

Definition 16. A seminorm on a vector space X is a function $p : X \to [0, \infty)$ such that:

1. $p(x + y) \leq p(x) + p(y)$ for all $x, y \in X$,
2. $p(\lambda x) = |\lambda| \, p(x)$ for all $\lambda \in \mathbb{K}$ and $x \in X$.

Definition 17 (Fréchet Space)**.** A *Fréchet space* is a topological vector space X satisfying:

1. X is **locally convex**.
2. Its topology is generated by a **countable family of seminorms** $\{p_n\}_{n \in \mathbb{N}}$.
3. X is **metrizable**, meaning there exists a metric inducing this topology.
4. X is **complete**, i.e. every Cauchy sequence converges in X.

Equivalently a Fréchet space is simply a *complete, metrizable, locally convex topological vector space.* Notice the only difference in the definition here and the definition we give of an F-space is that the latter does not need to be locally-convex. Some modern treatments also use the term F-space to denote what we are calling here a Fréchet space.

Metric definition Starting from a countable defining family of seminorms $\{p_n\}_{n\in\mathbb{N}}$, one can actually write down an explicit metric. A standard choice is

$$d(x,y) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{p_n(x-y)}{1+p_n(x-y)}.$$

This metric is translation-invariant and, although the formula may look a bit artificial at first glance, it induces exactly the given topology. Moreover, completeness of this metric is equivalent to completeness of the space itself.

Examples

1. **Banach spaces:** Any Banach space is automatically a Fréchet space, since a norm is just a single seminorm and already gives a complete metric.

2. **Spaces of smooth functions:** Let $\Omega \subseteq \mathbb{R}^n$ be open. The space

   $$C^\infty(\Omega)$$

   can be turned into a Fréchet space by using seminorms of the form

   $$p_{K,m}(f) = \sup_{x\in K} \max_{|\alpha|\leq m} |D^\alpha f(x)|,$$

   where $K \subset \Omega$ is compact and $m \in \mathbb{N}$. The dependence on both K and m is essential here, even if it makes the notation a bit heavier.

3. **Schwartz space:** The space $\mathcal{S}(\mathbb{R}^n)$ of rapidly decreasing smooth functions is another standard example. Its topology is given by seminorms

   $$p_{\alpha,\beta}(f) = \sup_{x\in\mathbb{R}^n} |x^\beta D^\alpha f(x)|,$$

   which encode both decay and differentiability in a single expression.

4. **Holomorphic functions:** The space of holomorphic functions on a domain $D \subset \mathbb{C}$, equipped with the topology of uniform convergence on compact subsets, also forms a Fréchet space. Here, the seminorms come from taking suprema over compact sets.

Remarks on Variants of the Definition

The terminology is not completely uniform across the literature:

- Some authors drop the requirement of local convexity; in that case, one often speaks instead of an *F-space*, as was done in the previous sections, we also make this distinction.
- In most modern treatments, however, local convexity is included, largely because it allows one to use tools such as the Hahn–Banach theorem without further complications.

7.3 Significance in Functional Analysis

Fréchet spaces occupy something like a middle ground in functional analysis. On the one hand, they are general enough to include:

- spaces of smooth or analytic functions,
- the setting for distribution theory,
- certain constructions in infinite-dimensional differential geometry,
- and modes of convergence that go beyond what norms alone can capture.

On the other hand, their completeness and metrizability enables them to still retain enough structure to support a reasonably well-behaved theory of linear operators, dual spaces, and convergence.

Fréchet spaces arise quite naturally once one tries to extend Banach space methods to broader classes of function spaces. Tracing back to Fréchet's early work on abstract metric structures, they represent an intermediate but very useful level of generality, where topological, algebraic, and analytic ideas all interact in a way that is not entirely rigid, but still far from arbitrary. All of these factors, make generalising frames to Fréchet spaces a very attractive prospect. The proceeding sections describe other ways to define a Fréchet space, equivalent formations which will make defining frames in them more natural and easier to see.

7.4 Fréchet Spaces through Projective Limits

7.4.1 Projective Topologies

The theory that we discuss below can be typically found in standard topological vector space references like Schafer and Robertson and Robertson [86, 81]. A systematic treatment to the theory of locally convex spaces and the projective and inductive limits was developed by Grothendieck and their contemporaries in the 1950s [44, 16, 87].

Let E and $\{E_\alpha\}_{\alpha\in A}$ be vector spaces over a field $\mathbb{K}$. For each $\alpha \in A$, let τ_α be a locally convex topology on E_α, and let

$$f_\alpha : E \to E_\alpha$$

be a linear mapping.

Definition 18 (Projective topology)**.** The *projective topology* τ on E with respect to the family $\{(E_\alpha, \tau_\alpha, f_\alpha)\}_{\alpha\in A}$ is defined as the *coarsest* topology on E for which each f_α is continuous as a map from (E, τ) into (E_α, τ_α).

Equivalently, τ is the least upper bound (in the lattice of all topologies on E) of the family of inverse image topologies

$$f_\alpha^{-1}(\tau_\alpha), \quad \alpha \in A.$$

A neighborhood base of a point $x \in E$ with respect to τ is given as follows: if we denote $x_\alpha = f_\alpha(x)$, then a base of neighborhoods of x consists of all sets of the form

$$\bigcap_{\alpha\in H} f_\alpha^{-1}(U_\alpha),$$

where $H \subseteq A$ is finite and for each $\alpha \in H$, U_α is a neighborhood of x_α in (E_α, τ_α).

Since each f_α is linear and each (E_α, τ_α) is locally convex, it follows that τ is translation-invariant and admits a base of convex neighborhoods of 0 satisfying the axioms of a locally convex space. Hence (E, τ) is a locally convex space.

The projective topology τ on E is Hausdorff if and only if for every $x \in E$, $x \neq 0$, there exists $\alpha \in A$ and a 0-neighborhood U_α in E_α such that

$$f_\alpha(x) \notin U_\alpha.$$

Theorem 19. *Let F be a topological space and let $u : F \to E$ be a mapping. Then u is continuous with respect to the projective topology τ on E if and only if for every $\alpha \in A$, the composition*

$$f_\alpha \circ u : F \to E_\alpha$$

is continuous.

Proof. Let τ denote the projective topology on E with respect to the family

$$\{(E_\alpha, \tau_\alpha, f_\alpha)\}_{\alpha \in A}.$$

By definition, τ is the coarsest topology on E for which every map

$$f_\alpha : (E, \tau) \to (E_\alpha, \tau_\alpha)$$

is continuous.

We prove the two implications separately.

Assume that $u : F \to E$ is continuous. Since each $f_\alpha : (E, \tau) \to (E_\alpha, \tau_\alpha)$ is continuous by the definition of τ, the composition of continuous maps

$$f_\alpha \circ u : F \longrightarrow E_\alpha$$

is continuous for every $\alpha \in A$.

Assume now that for every $\alpha \in A$, the map $f_\alpha \circ u : F \to E_\alpha$ is continuous. We must show that $u : F \to (E, \tau)$ is continuous.

Let $V \subseteq E$ be open in the projective topology τ. Since τ is the initial topology induced by the family $\{f_\alpha\}_{\alpha \in A}$, every open set V is a union of finite intersections of sets of the form $f_\alpha^{-1}(U_\alpha)$, where each U_α is open in E_α. Thus it suffices to prove that

$$u^{-1}\left(\bigcap_{\alpha \in H} f_\alpha^{-1}(U_\alpha)\right)$$

is open in F for every finite subset $H \subseteq A$ and every choice of open sets $U_\alpha \subseteq E_\alpha$.

But

$$u^{-1}\left(\bigcap_{\alpha \in H} f_\alpha^{-1}(U_\alpha)\right) = \bigcap_{\alpha \in H} u^{-1}\big(f_\alpha^{-1}(U_\alpha)\big) = \bigcap_{\alpha \in H} (f_\alpha \circ u)^{-1}(U_\alpha).$$

By hypothesis, each composition $f_\alpha \circ u$ is continuous, so each set

$$(f_\alpha \circ u)^{-1}(U_\alpha)$$

is open in F. Since finite intersections of open sets are open, the displayed set is open in F.

Therefore $u^{-1}(V)$ is open for every open set $V \subseteq E$, and hence u is continuous.

This proves that $u : F \to E$ is continuous if and only if each $f_\alpha \circ u : F \to E_\alpha$ is continuous. □

Examples

(a) Subspaces. If M is a subspace of a locally convex space (E, τ), then the induced topology on M is precisely the projective topology with respect to the canonical embedding $M \hookrightarrow E$.

(b) Products. Let $\{(E_\alpha, \tau_\alpha)\}_{\alpha \in A}$ be a family of locally convex spaces. The product space

$$E = \prod_{\alpha \in A} E_\alpha$$

with the product topology is locally convex, and this topology coincides with the projective topology with respect to the canonical projections

$$p_\alpha : E \to E_\alpha.$$

(c) Upper Bounds. If $\{\tau_\alpha\}_{\alpha \in A}$ is a family of locally convex topologies on a vector space E, then their least upper bound τ is again locally convex and coincides with the projective topology induced by the identity maps

$$(E, \tau_\alpha, \mathrm{id}) \quad (\alpha \in A).$$

(d) Weak Topologies. Let E be a vector space and let $F \subseteq E^*$ be a nonempty subset. The weak topology $\sigma(E, F)$ is the projective topology induced by the family of maps

$$f : E \to \mathbb{K}, \quad f \in F.$$

This topology is locally convex if and only if F separates points of E.

7.4.2 Projective Limits

Let $(A, \leq)$ be a directed set, and let $\{(E_\alpha, \tau_\alpha)\}_{\alpha \in A}$ be a family of locally convex spaces. Suppose that for each pair $\alpha \leq \beta$, there exists a continuous linear map

$$g_{\alpha\beta} : E_\beta \to E_\alpha,$$

such that:

- $g_{\alpha\alpha} = \mathrm{id}_{E_\alpha}$,
- $g_{\alpha\beta} \circ g_{\beta\gamma} = g_{\alpha\gamma}$ whenever $\alpha \leq \beta \leq \gamma$.

Definition 20 (Projective limits)**.** Define

$$F = \prod_{\alpha \in A} E_\alpha,$$

and let

$$E = \{x = (x_\alpha) \in F : x_\alpha = g_{\alpha\beta}(x_\beta) \text{ whenever } \alpha \leq \beta\}.$$

Then E is called the *projective limit* of the system $\{E_\alpha, g_{\alpha\beta}\}$ and is denoted

$$E = \varprojlim E_\alpha.$$

The topology on E is the subspace topology inherited from F, which coincides with the projective topology with respect to the maps

$$f_\alpha : E \to E_\alpha, \quad f_\alpha(x) = x_\alpha.$$

If each (E_α, τ_α) is complete (respectively quasi-complete), then the product $F = \prod_\alpha E_\alpha$ is complete (respectively quasi-complete). Since

$$E = \bigcap_{\alpha \leq \beta} V_{\alpha\beta}, \quad V_{\alpha\beta} = \{x \in F : x_\alpha - g_{\alpha\beta}(x_\beta) = 0\},$$

and each $V_{\alpha\beta}$ is closed in F, it follows that E is closed in F. Hence E is complete (respectively quasi-complete).

Theorem 21 ([86])**.** *Every Fréchet space is topologically isomorphic to a projective limit of a sequence of Banach spaces.*

Proof. Let E be a Fréchet space. By definition, E is a complete, metrizable, locally convex topological vector space. Hence its topology is generated by a countable family of continuous seminorms. Choose such a family and, after replacing it by an equivalent increasing family if necessary, we may assume that

$$p_1 \leq p_2 \leq p_3 \leq \cdots$$

and that the topology of E is the initial topology induced by the seminorms $(p_n)_{n\in\mathbb{N}}$.

For example, if $(q_n)_{n\in\mathbb{N}}$ is any countable family of seminorms generating the topology, one may define

$$p_n := q_1 + q_2 + \cdots + q_n,$$

so that (p_n) is increasing and still generates the same topology.

For each $n \in \mathbb{N}$, let

$$N_n := \ker p_n = \{x \in E : p_n(x) = 0\},$$

and consider the quotient space

$$E_n^0 := E/N_n.$$

Define a norm on E_n^0 by

$$\|x + N_n\|_n := p_n(x), \qquad x \in E.$$

This is well defined because if $x - y \in N_n$, then $p_n(x - y) = 0$, and hence $p_n(x) = p_n(y)$.

Let $\widehat{E}_n$ denote the completion of the normed space $(E_n^0, \|\cdot\|_n)$. Then $\widehat{E}_n$ is a Banach space.

Because $p_n \leq p_{n+1}$, we have

$$N_{n+1} \subseteq N_n.$$

Therefore the canonical quotient map

$$j_{n,n+1} : E_{n+1}^0 \to E_n^0, \qquad j_{n,n+1}(x + N_{n+1}) = x + N_n$$

is well defined. Moreover,

$$\|j_{n,n+1}(x + N_{n+1})\|_n = p_n(x) \leq p_{n+1}(x) = \|x + N_{n+1}\|_{n+1},$$

so $j_{n,n+1}$ is continuous and has operator norm at most 1.

Since E_n^0 is dense in $\widehat{E}_n$, each $j_{n,n+1}$ extends uniquely to a continuous linear map

$$g_{n,n+1} : \widehat{E}_{n+1} \to \widehat{E}_n.$$

For $m > n$, define

$$g_{n,m} := g_{n,n+1} \circ g_{n+1,n+2} \circ \cdots \circ g_{m-1,m}.$$

The family $\big(\widehat{E}_n, g_{n,m}\big)$ is thus a projective system of Banach spaces.

Let

$$X := \varprojlim \widehat{E}_n = \left\{ (x_n)_{n\in\mathbb{N}} \in \prod_{n\in\mathbb{N}} \widehat{E}_n : g_{n,m}(x_m) = x_n \text{ whenever } n \le m \right\}.$$

Since each $\widehat{E}_n$ is Banach, the product $\prod_n \widehat{E}_n$ is complete, and X is a closed subspace of that product. Hence X is complete.

For each n, let

$$\iota_n : E \to \widehat{E}_n$$

be the composition of the quotient map $E \to E_n^0$ with the canonical isometric embedding $E_n^0 \hookrightarrow \widehat{E}_n$. Define

$$T : E \to X, \qquad T(x) := \big(\iota_n(x)\big)_{n\in\mathbb{N}}.$$

This map is well defined because the family $\big(\iota_n(x)\big)$ is compatible with the connecting maps: for $n \le m$,

$$g_{n,m}\big(\iota_m(x)\big) = \iota_n(x).$$

The map T is linear and continuous.

Suppose that $T(x) = 0$. Then $\iota_n(x) = 0$ for every n, so $p_n(x) = 0$ for every n. Since the seminorms (p_n) generate the topology of the Hausdorff space E, they separate points; hence $x = 0$. Therefore T is injective.

We now show that the topology of E coincides with the subspace topology induced by T from X.

Indeed, for each n, if $\pi_n : X \to \widehat{E}_n$ denotes the canonical projection, then

$$\pi_n \circ T = \iota_n.$$

Moreover, for every $x \in E$,

$$p_n(x) = \|\iota_n(x)\|_{\widehat{E}_n} = \|\pi_n(T(x))\|_{\widehat{E}_n}.$$

Thus the seminorms that define the topology of E are exactly the pullbacks under T of the norm topologies of the coordinate Banach spaces. Since the projective limit topology on X is the initial topology induced by the family $(\pi_n)_{n\in\mathbb{N}}$, it follows that the topology on E is precisely the topology transported from $T(E) \subseteq X$ by T.

Consequently, T is a homeomorphism of E onto its image $T(E)$.

We prove that $T(E)$ is dense in X.

Let $y = (y_n) \in X$, and let U be a basic neighborhood of y in X. By the definition of the product topology, there exist an index N and neighborhoods $V_1, \dots, V_N$ with V_n a neighborhood of y_n in $\widehat{E}_n$ such that

$$U = X \cap \bigcap_{n=1}^{N} \pi_n^{-1}(V_n).$$

Because the connecting maps are continuous, for each $n \leq N$ there exists a neighborhood W of y_N in $\widehat{E}_N$ such that

$$g_{n,N}(W) \subseteq V_n \qquad \text{for every } n = 1, \dots, N.$$

Now E_N^0 is dense in $\widehat{E}_N$, so there exists $x \in E$ such that $\iota_N(x) \in W$.

For this x, and for each $n \leq N$, compatibility of the system gives

$$\pi_n(T(x)) = \iota_n(x) = g_{n,N}\big(\iota_N(x)\big) \in V_n.$$

Hence $T(x) \in U$. Since every basic neighborhood of every point of X meets $T(E)$, the set $T(E)$ is dense in X.

The space E is complete, and T is a homeomorphism of E onto $T(E)$. Therefore $T(E)$ is complete in the subspace topology inherited from X.

Since X is Hausdorff and complete, every complete subspace of X is closed. Hence $T(E)$ is closed in X.

But we have already proved that $T(E)$ is dense in X. Therefore $T(E)$ is both dense and closed in X, so

$$T(E) = X.$$

Thus T is surjective.

The map

$$T : E \to \varprojlim \widehat{E}_n$$

is linear, bijective, continuous, and has continuous inverse. Hence it is a topological isomorphism.

Therefore every Fréchet space is topologically isomorphic to a projective limit of a sequence of Banach spaces. □

Thus every Fréchet space can be realized as a projective limit of Banach spaces, preserving both linear and topological structure. This representation is fundamental in reducing problems in Fréchet spaces to problems in Banach spaces.

8 Generalising Frame Theory from Hilbert and Banach Spaces

We will now discuss a body of work primarily authored by Pilipović and Stoeva [75, 78, 76, 77, 92, 79, 8, 91, **?**, 1]. The primary object of this work is to define frames in the general setting of Fréchet spaces through projective limits of infinite sequences of semi-norms and Banach spaces. We will also discuss some specific cases where these frames are defined on Schwartz spaces and a special generalised version of GF-frames which are defined on Gelfand-Shilov spaces.

To build a natural progression towards the complex topological structures of Fréchet spaces, it is imperative to first establish the foundational concepts of standard frame theory. In this section we, once again, detail the transition from the highly structured environment of Hilbert spaces to the more general setting of Banach spaces, where the standard sequence space ℓ^2 is replaced by general sequence spaces governed by specific topological properties.

The theory of frames was initially developed to address the need for stable, redundant series expansions in Hilbert spaces, offering a more flexible alternative to the strict uniqueness required by orthonormal bases.

Definition 22 (Frame for a Hilber Space)**.** Let $\mathcal{H}$ be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\| \cdot \|_{\mathcal{H}}$. A countable sequence $(e_n)_{n=1}^{\infty}$ with elements from $\mathcal{H}$ is called a *frame* for $\mathcal{H}$ if there exist positive constants A and B (called the lower and upper frame bounds, respectively) satisfying the fundamental frame inequalities:

$$A\|x\|_{\mathcal{H}}^2 \leq \sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2 \leq B\|x\|_{\mathcal{H}}^2, \quad \text{for all } x \in \mathcal{H}. \tag{2}$$

When analyzing these representations, the associated operators play a critical role. Given a frame $E = (e_n)_{n=1}^{\infty}$, the *frame operator* $S_E : \mathcal{H} \to \mathcal{H}$ is determined by the composition of the analysis and synthesis processes:

$$S_E x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n. \tag{3}$$

A pivotal result in standard frame theory is that S_E constitutes a bounded bijection from $\mathcal{H}$ onto $\mathcal{H}$ [22]. Because S_E is invertible, one can stably reconstruct any signal $x \in \mathcal{H}$ using the inverse frame operator S_E^{-1}.

This leads to the concept of dual frames. While a *Riesz basis* (a frame that is simultaneously a Schauder basis) offers unique representations and possesses exactly one dual frame [80, 11, 90], an *overcomplete* or *redundant* frame (which is not a Schauder basis) offers infinitely many dual frames for signal reconstruction. A sequence $(u_n)_{n=1}^{\infty} \subset \mathcal{H}$ is called a *dual frame* of $(e_n)_{n=1}^{\infty}$ if it facilitates the expansion:

$$x = \sum_{n=1}^{\infty} \langle x, e_n \rangle u_n = \sum_{n=1}^{\infty} \langle x, u_n \rangle e_n, \quad \text{for all } x \in \mathcal{H}. \tag{4}$$

Among these infinitely many duals, the sequence $(S_E^{-1} e_n)_{n=1}^{\infty}$ is mathematically distinguished. It is defined as the *canonical dual frame* of E and is typically denoted as $(\tilde{e}_n)_{n=1}^{\infty}$ [22]. While the canonical dual possesses highly desirable properties such as minimizing the ℓ^2-norm of the expansion coefficients, it may lack other application-specific characteristics, such as compact support or optimal smoothness. Thus, the redundancy of frames and the availability of alternative, non-canonical dual frames are highly advantageous.

8.1 Transition to Banach Spaces

Generalizing frame theory to a Banach space X necessitates replacing the standard ℓ^2 coefficient space with a broader class of sequence spaces. We denote a general Banach sequence space by $(\Theta, |||\cdot|||_\Theta)$. To ensure that coordinate analysis behaves continuously and that basis expansions are well-defined, Θ must exhibit specific structural properties [41, 5].

Definition 23 (Solid sequence space)**.** A sequence space Θ is considered *solid* if the following domination property holds: if a sequence $\{c_i\}_{i=1}^{\infty} \in \Theta$ and a scalar sequence $\{d_i\}_{i=1}^{\infty}$ satisfies $|d_i| \leq |c_i|$ for all $i \in \mathbb{N}$, then $\{d_i\}_{i=1}^{\infty} \in \Theta$ and the norm inequality $|||\{d_i\}|||_\Theta \leq |||\{c_i\}|||_\Theta$ is satisfied [75].

Definition 24 (BK-Space and CB-space, [76, 78])**.** Let Θ be a Banach sequence space.

- Θ is a **BK-space** if all of its coordinate functionals are continuous. This implies that convergence within the space Θ strictly guarantees the convergence of the individual sequence coordinates.

- A BK-space Θ is termed a **λ-BK-space** (where $\lambda \geq 1$) if it contains all canonical unit vectors $e_i = \{\delta_{ki}\}_{k=1}^{\infty}$ and satisfies the uniform boundedness condition for partial sums:

$$\left|\left|\left|\sum_{i=1}^{n} c_i e_i\right|\right|\right|_{\Theta} \leq \lambda |||\{c_i\}_{i=1}^{\infty}|||_{\Theta}, \quad \text{for all } n \in \mathbb{N} \text{ and } \{c_i\}_{i=1}^{\infty} \in \Theta. \quad (5)$$

- A BK-space for which the canonical unit vectors $\{e_i\}_{i=1}^{\infty}$ form a Schauder basis is called a **CB-space**.

In a CB-space, the canonical basis constant is determined by $\sup_N \|S_N\| < \infty$, where the truncation operator $S_N : \Theta \to \Theta$ is defined by $S_N(\{c_i\}) = \sum_{i=1}^{N} c_i e_i$. By definition, every CB-space inherently functions as a λ-BK-space, where the constant λ coincides with the canonical basis constant. Prominent examples of 1-BK-spaces include solid CB-spaces such as the Lebesgue sequence spaces ℓ^p $(1 \leq p < \infty)$ and c_0 [46].

Proposition 25 (Duals of CB-spaces, [75])**.** *If Θ is a CB-space, the set $BK\Theta^* := \{\{g(e_i)\}_{i=1}^{\infty} : g \in \Theta^*\}$ equipped with the norm $|||\{g(e_i)\}|||_{BK\Theta^*} := \|g\|_{\Theta^*}$ constitutes a BK-space that is isometrically isomorphic to the dual space Θ^* [4]. Furthermore, if Θ is a reflexive CB-space, the sequence of coefficient functionals associated with the canonical basis forms a Schauder basis for Θ^*, implying that $BK\Theta^*$ is also a CB-space.*

For rigorous theoretical development, when Θ is a CB-space, its dual Θ^* is universally identified with $BK\Theta^*$.

8.2 Θ-Frames and Banach Frames

With the sequence space framework established, the definition of a frame in a Banach space $(X, \|\cdot\|_X)$ is rigorously formulated relative to a chosen BK-space Θ.

Definition 26 (Θ-Frame and Θ-Bessel Sequence)**.** Let $(X, \|\cdot\|_X)$ be a Banach space, and let $(\Theta, |||\cdot|||_{\Theta})$ be a BK-space. A countable family of continuous linear functionals $\{g_i\}_{i\in I} \subset X^*$ is defined as a **Θ-frame** for X if there exist constants $0 < A \leq B < \infty$ such that the sequence of evaluations $\{g_i(f)\}_{i\in I}$ belongs to Θ and the following fundamental frame inequalities are satisfied:

$$A\|f\|_X \leq |||\{g_i(f)\}_{i\in I}|||_{\Theta} \leq B\|f\|_X, \quad \text{for all } f \in X \text{ [10, 11]}. \quad (6)$$

The constants A and B are referred to as the Θ-frame bounds. If the mapping $\{g_i(f)\}_{i\in I} \in \Theta$ is valid and only the upper inequality $|||\{g_i(f)\}|||_\Theta \leq B\|f\|_X$ holds for all $f \in X$, the sequence $\{g_i\}$ is termed a **Θ-Bessel sequence** for X with bound B. Note that if Θ is a BK-space, verifying that $\{g_i\}_{i=1}^\infty$ is a Θ-Bessel sequence naturally implies that every individual functional g_i must be bounded on X [77, 23].

While the Θ-frame condition bounds the sequence space norm above and below by the Banach space norm (establishing a topological isomorphism between X and the range of the analysis operator), it does not automatically guarantee that elements can be reconstructed from their coefficients via a bounded linear mapping. Elevating a Θ-frame to a full Banach frame requires the existence of a continuous reconstruction mechanism.

Definition 27 (Banach frame)**.** A Θ-frame $\{g_i\}_{i\in I}$ is elevated to the status of a **Banach frame** for X with respect to Θ if there exists a bounded linear reconstruction operator $S : \Theta \to X$ such that:

$$S(\{g_i(f)\}_{i\in I}) = f, \quad \text{for all } f \in X \text{ [3].} \tag{7}$$

Equivalently, defining the analysis operator $U : X \to \Theta$ by $Uf = \{g_i(f)\}_{i\in I}$, the sequence $\{g_i\}$ constitutes a Banach frame if and only if U possesses a bounded left inverse S defined on the entire sequence space Θ.

The transition from a Θ-frame to a Banach frame is dictated entirely by the geometrical disposition of the analysis operator's range within the sequence space Θ. Recall that a closed subspace X_1 of X is *complemented* in X if there exists a closed subspace X_2 such that $X_1 \cap X_2 = \{0\}$ and $X_1 + X_2 = X$, equivalent to the existence of a continuous projection from X onto X_1 along X_2.

Proposition 28 (Complemented Range Condition, [20])**.** *Let Θ be a BK-space and $\{g_i\} \subset X^*$ be a Θ-frame for X. The family $\{g_i\}$ is a Banach frame for X with respect to Θ if and only if the set $\{\{g_i(f)\} : f \in X\}$, which is the range of the analysis operator, is a complemented subspace in Θ.*

When the target sequence space possesses a canonical basis, this complementedness is intrinsically tied to the existence of dual series expansions.

Theorem 29 (Dual Expansions, [20])**.** *Let Θ be a CB-space and $\{g_i\} \subset X^*$ be a Θ-frame for X. There exists a Θ^*-Bessel sequence $\{f_i\} \subset X \subset X^{**}$ for*

X^* *such that the series expansion*

$$f = \sum_{i=1}^{\infty} g_i(f) f_i, \quad \textit{for all } f \in X \tag{8}$$

is valid if and only if the set $\{\{g_i(f)\} : f \in X\}$ *is complemented in* Θ *[16, 17]. Furthermore, if both* Θ *and* Θ^* *are CB-spaces, the complementedness of the range is the necessary and sufficient condition to ensure the existence of a sequence* $\{f_i\}$ *serving as a dual frame that simultaneously yields the dual expansion:*

$$g = \sum_{i=1}^{\infty} g(f_i) g_i, \quad \textit{for all } g \in X^* \tag{9}$$

By anchoring these principles, the theory demonstrates that redundant signal decomposition in Banach spaces explicitly requires controlling both the uniform topology of the coordinate functionals (via BK/CB-spaces) and the continuous projection mechanisms (complemented subspaces) within those sequence spaces. This baseline is the theoretical prerequisite for progressing to infinite families of nested localized norms, driving the definition of frames in Fréchet topologies.

9 Sequence Space Constructions and Density Theorems

Before advancing to the infinite families of nested norms that characterize Fréchet topologies, we must establish the mathematical tools that allow Banach frames to be constructed, extended, and verified efficiently. A fundamental problem in applied frame theory is determining whether frame conditions hold across an entire space. Often, verifying these bounds on a dense subset is significantly more tractable. If X is a Hilbert space and $\Theta = \ell^2$, it is well known that if the frame bounds hold on a dense subset of X, they naturally extend to all of X. The same conclusion holds when X is a Banach space and the sequence space is ℓ^p for $p \in (1, \infty)$. This section generalizes these density arguments to arbitrary λ-BK-spaces and addresses the profound challenge of constructing tailored sequence spaces for sequences that do not natively act as Bessel sequences [75, 76, 21, 61, 66].

9.1 Density Extensions for Θ-Frames

To provide a robust foundation for general Banach frames, we extend the density argument to the class of λ-BK-spaces. The following theorem demonstrates that if the frame inequalities and sequence space containment hold on a dense subset of a Banach space, they continuously extend to the entirety of the space, albeit with the upper bound scaled by the structural constant λ of the BK-space.

Theorem 30 (Density Theorem for Θ-Frames, [75])**.** *Let $(\Theta, |||\cdot|||)$ be a λ-BK-space. Let W be a dense subset of a Banach space $(X, \|\cdot\|)$ and let $\{g_i\} \in (X^*)^{\mathbb{N}}$.*

(i) If $\{g_i(f)\} \in \Theta$ and the upper inequality $|||\{g_i(f)\}||| \leq B\|f\|_X$ hold for all $f \in W$, then $\{g_i\}$ is a Θ-Bessel sequence for X with a bound λB.

(ii) If both the lower and upper Θ-frame inequalities hold for all $f \in W$ (i.e., $A\|f\|_X \leq |||\{g_i(f)\}||| \leq B\|f\|_X$), then $\{g_i\}$ is a Θ-frame for X with bounds A and λB.

Proof. We begin by proving part (i). Assume that $\{g_i\}$ satisfies the containment condition and the upper inequality for all $f \in W$. We must first prove that $\{g_i(f)\} \in \Theta$ for all $f \in X$. Fix an arbitrary $f \in X \setminus W$ and assume, for the sake of contradiction, that there exists an $N \in \mathbb{N}$ such that $|||\sum_{i=1}^{N} g_i(f)e_i||| > \lambda B\|f\|$. We can then define a strictly positive value $\epsilon := |||\sum_{i=1}^{N} g_i(f)e_i||| - \lambda B\|f\| > 0$.

By the density of W in X, there exists a sequence $\{f_n\} \in W^{\mathbb{N}}$ such that $f_n \to f$ as $n \to \infty$. Consequently, there exists an $N_1 \in \mathbb{N}$ such that for all $n > N_1$, we have $0 \leq \|f_n\| - \|f\| < \frac{\epsilon}{2\lambda B}$. Furthermore, due to the continuity of the finite number of linear functionals g_i (for $i = 1, 2, \ldots, N$), there exists an $N_2 \in \mathbb{N}$ such that $|||\sum_{i=1}^{N} g_i(f_n - f)e_i||| < \frac{\epsilon}{2}$ for all $n > N_2$.

For any $n > \max\{N_1, N_2\}$, we can apply the triangle inequality to derive:

$$\begin{aligned}\left|\left|\left|\sum_{i=1}^{N} g_i(f_n)e_i\right|\right|\right| &> \lambda B\|f\| + \epsilon - \frac{\epsilon}{2} \\ &> \lambda B\left(\|f_n\| - \frac{\epsilon}{2\lambda B}\right) + \frac{\epsilon}{2} = \lambda B\|f_n\|.\end{aligned}$$

Because Θ is a λ-BK-space, the uniform boundedness condition for partial sums dictates that $|||\{g_i(f_n)\}_{i=1}^{\infty}||| \geq \frac{1}{\lambda}|||\sum_{i=1}^{N} g_i(f_n)e_i|||$. Combining

this with our previous inequality yields $|||\{g_i(f_n)\}_{i=1}^{\infty}||| > B\|f_n\|$, which strictly contradicts the assumption that the upper bound B holds for $f_n \in W$. Therefore, our initial assumption must be false, and it must hold that $|||\sum_{i=1}^{N} g_i(f)e_i||| \leq \lambda B\|f\|$ for all $N \in \mathbb{N}$.

Thus, for every $N \in \mathbb{N}$, the partial sum operator $S_N : X \to \Theta$, defined by $S_N(f) = \sum_{i=1}^{N} g_i(f)e_i$, is a bounded linear operator satisfying $\|S_N\| \leq \lambda B$. Moreover, $S_N(f)$ converges as $N \to \infty$ for all f in the dense subset W of X. By standard operator theory, a uniformly bounded sequence of operators converging on a dense subset must converge everywhere on the space, meaning $S_N(f)$ converges for all $f \in X$. Because Θ is a BK-space, coordinate convergence is guaranteed, meaning the limit $\lim_{N\to\infty} S_N(f)$ is exactly the sequence $\{g_i(f)\}_{i=1}^{\infty}$. Hence, $\{g_i(f)\}_{i=1}^{\infty} \in \Theta$ for all $f \in X$. The Banach-Steinhaus theorem (Uniform Boundedness Principle) then implies that $|||\{g_i(f)\}_{i=1}^{\infty}||| \leq \lambda B\|f\|$ for all $f \in X$, establishing $\{g_i\}$ as a Θ-Bessel sequence for X with bound λB.

To prove part (ii), assume that both the lower and upper bounds hold for all $f \in W$. It remains only to verify the validity of the lower Θ-frame inequality for all $f \in X \setminus W$. Take $f \in X \setminus W$ and a sequence $\{f_n\} \in W^{\mathbb{N}}$ such that $f_n \to f$ as $n \to \infty$. For any $n \in \mathbb{N}$, the sequence elements satisfy $|||\{g_i(f_n - f)\}_{i=1}^{\infty}||| \leq \lambda B\|f_n - f\|$. Taking the limit as $n \to \infty$, we get $\lim_{n\to\infty} |||\{g_i(f_n)\}_{i=1}^{\infty}||| = |||\{g_i(f)\}_{i=1}^{\infty}|||$. By the assumption on W, the inequality $A\|f_n\| \leq |||\{g_i(f_n)\}_{i=1}^{\infty}|||$ holds for all $n \in \mathbb{N}$. When we take the limit as $n \to \infty$, we obtain $A\|f\| \leq |||\{g_i(f)\}_{i=1}^{\infty}|||$, which completes the proof. □

9.2 Constructive Methodologies for Sequence Spaces

A significant limitation of standard Banach frame theory is the assumption that the given sequence $\{g_i\}$ acts natively as a Bessel sequence with respect to a known sequence space Θ. In practice, we encounter sequences that generate valid series expansions but fail to be Bessel sequences for standard spaces. For example, given an orthonormal basis $\{e_i\}_{i=1}^{\infty}$ for a Hilbert space H, the sequence $\{g_i^1\}_{i=1}^{\infty} := \{e_1, e_1, 2e_2, 3e_3, 4e_4, \dots\}$ is not a Hilbert frame, nor a Bessel sequence with respect to ℓ^2. However, valid series expansions of the form $f = \sum_{i=1}^{\infty} g_i(f)f_i$ still exist via the dual sequence $\{f_i^1\}_{i=1}^{\infty} := \{e_1, 0, \frac{1}{2}e_2, \frac{1}{3}e_3, \frac{1}{4}e_4, \dots\}$.

To mathematically accommodate such sequences, we can systematically construct a new sequence space $\widetilde{\Theta}$ from an initial space Θ such that the given

sequence $\{g_i\}$ successfully functions as a Banach frame with respect to this constructed space.

Definition 31 (The Space $\tilde{\Theta}$)**.** Let $(X, \|\cdot\|) \neq \{0\}$ be a reflexive Banach space, and let $\{g_i\}_{i=1}^{\infty} \in (X^*)^{\mathbb{N}} \setminus \{0\}$. For every scalar sequence $c = \{c_i\}_{i=1}^{\infty}$, we define the localized evaluation set M_c as:

$$M_c := \{f \in X : |c_i| \leq |g_i(f)|, \forall i \in \mathbb{N}\}. \tag{10}$$

We define the tailored sequence space $\tilde{\Theta}$ as the set of all sequences for which this set is non-empty, equipped with a norm defined by the infimum over M_c:

$$\tilde{\Theta} := \{c = \{c_i\}_{i=1}^{\infty} : M_c \neq \emptyset\}, \quad |||c|||_{\tilde{\Theta}} := \inf\{\|f\| : f \in M_c\}. \tag{11}$$

To ensure that $\tilde{\Theta}$ possesses the topological properties required for frame analysis (namely, being a solid CB-space), we introduce three axiomatic conditions.

- **(A1):** $(\forall c \in \tilde{\Theta})(\forall d \in \tilde{\Theta})(\forall f \in M_c)(\forall h \in M_d) \implies (\exists r \in M_{c+d})(\|r\| \leq \|f\| + \|h\|)$. This condition ensures the triangle inequality holds for the constructed norm $|||\cdot|||_{\tilde{\Theta}}$.
- **(A2):** $(\forall c \in \tilde{\Theta})(\forall \epsilon > 0)(\exists k \in \mathbb{N})(\exists f \in M_{c(k)})(\|f\| < \epsilon)$, where $c(k)$ is the sequence c with its first k elements replaced by 0. This condition forces the tail of the sequences to decay in norm, serving as the necessary and sufficient condition for the canonical unit vectors to form a Schauder basis (the CB-space property).
- **(A3):** There exists $A \in (0, 1]$ such that for every $f \in X$, the condition $\tilde{f} \in M_{\{g_i(f)\}_{i=1}^{\infty}}$ implies $A\|f\| \leq \|\tilde{f}\|$. This serves as the generalized lower frame bound condition.

Theorem 32 (Properties of the Constructed Space)**.** *Assume that condition (A1) is fulfilled. Then the following statements hold:*

(a) $\tilde{\Theta}$ *is a solid BK-space and* $\{g_i\}_{i=1}^{\infty}$ *is a* $\tilde{\Theta}$*-Bessel sequence for* X *with bound* $\tilde{B} = 1$*.*

(b) $\{g_i\}_{i=1}^{\infty}$ *is a* $\tilde{\Theta}$*-frame for* X *if and only if (A3) holds, which is equivalent to the existence of a solid BK-space* $\Theta \supseteq \tilde{\Theta}$ *such that* $\{g_i\}_{i=1}^{\infty}$ *is a* Θ*-frame for* X*.*

(c) $\{g_i\}_{i=1}^{\infty}$ *is a Banach frame for* X *with respect to* $\tilde{\Theta}$ *if and only if there exists a solid BK-space* $\Theta \supseteq \tilde{\Theta}$ *such that* $\{g_i\}_{i=1}^{\infty}$ *is a Banach frame for* X *with respect to* Θ.

(d) $\tilde{\Theta}$ *is a CB-space if and only if (A2) holds.*

Proof. (a) To prove that $\tilde{\Theta}$ is a vector space and $|||\cdot|||_{\tilde{\Theta}}$ is a norm, we rely heavily on (A1). First, we show that $|||c|||_{\tilde{\Theta}} = 0$ implies $c_i = 0$ for all $i \in \mathbb{N}$. Let $|||c|||_{\tilde{\Theta}} = 0$. Fix $i \in \mathbb{N}$. For every $\epsilon > 0$, the infimum definition guarantees there exists $f_\epsilon \in M_c$ such that $\|f_\epsilon\| < \epsilon/\|g_i\|$. Consequently, $|c_i| \leq |g_i(f_\epsilon)| \leq \|g_i\|\|f_\epsilon\| < \epsilon$, which implies that $c_i = 0$.

For the solidity property, let $c = \{c_i\}_{i=1}^{\infty} \in \tilde{\Theta}$ and let $d = \{d_i\}_{i=1}^{\infty}$ be a sequence such that $|d_i| \leq |c_i|$ for all $i \in \mathbb{N}$. Since $M_d \supseteq M_c \neq \emptyset$, it naturally follows that $d \in \tilde{\Theta}$ and $|||d|||_{\tilde{\Theta}} \leq |||c|||_{\tilde{\Theta}}$.

To prove completeness, we first establish that the coordinate functionals are continuous. Fix $i \in \mathbb{N}$. If $c \in \tilde{\Theta}$, then for every $f \in M_c$, we have $|c_i| \leq |g_i(f)| \leq \|g_i\|\|f\|$. Taking the infimum over all $f \in M_c$ yields $|c_i| \leq \|g_i\| \inf\{\|f\| : f \in M_c\} = \|g_i\|||c|||_{\tilde{\Theta}}$.

Let $c^\nu = \{c_i^\nu\}_{i=1}^{\infty}$, $\nu \in \mathbb{N}$, be a Cauchy sequence in $\tilde{\Theta}$. Fix an arbitrary $\epsilon > 0$. There exists $\nu_0(\epsilon)$ such that for all $\mu \geq \nu_0$ and $\nu \geq \nu_0$, there exists an element $f_{\mu,\nu} \in X$ such that $\|f_{\mu,\nu}\| < \epsilon$ and $|c_i^\mu - c_i^\nu| \leq |g_i(f_{\mu,\nu})|$ for all $i \in \mathbb{N}$. Because the coordinate functionals are continuous, for every $i \in \mathbb{N}$, the sequence c_i^ν is a Cauchy sequence in $\mathbb{R}$ and converges to some number c_i as $\nu \to \infty$. Let $c := \{c_i\}_{i=1}^{\infty}$. Since $\|f_{\mu,\nu}\| < \epsilon$ for every $\mu \geq \nu_0$, there exists a bounded convergent subsequence of norms. Because X is a reflexive Banach space and the sequence $\{f_{\mu_k,\nu}\}_{k=1}^{\infty}$ is norm-bounded, the Eberlein-Šmulian theorem guarantees the existence of a subsequence $\{f_{\mu_{k_n},\nu}\}_{n=1}^{\infty}$ that converges weakly to some element $F^\nu \in X$. The weak lower semicontinuity of the norm implies $\|F^\nu\| \leq \liminf \|f_{\mu_{k_n},\nu}\| \leq \epsilon$. For every $i \in \mathbb{N}$, we have $|c_i - c_i^\nu| \leq |g_i(F^\nu)|$, establishing that $F^\nu \in M_{c-c^\nu}$. This confirms that $c - c^\nu \in \tilde{\Theta}$, so $c \in \tilde{\Theta}$, and $|||c - c^\nu|||_{\tilde{\Theta}} \leq \|F^\nu\| \leq \epsilon$ for all $\nu \geq \nu_0$, proving completeness. For every $f \in X$, the set $M_{\{g_i(f)\}}$ naturally contains f, so $\{g_i(f)\} \in \tilde{\Theta}$ and $|||\{g_i(f)\}|||_{\tilde{\Theta}} \leq \|f\|$, making $\{g_i\}$ a $\tilde{\Theta}$-Bessel sequence with bound $\tilde{B} = 1$.

(b) Assume (A3) holds. By definition, $A\|f\| \leq \inf\{\|\tilde{f}\| : \tilde{f} \in M_{\{g_i(f)\}}\} = |||\{g_i(f)\}|||_{\tilde{\Theta}}$ for all $f \in X$. Combined with (a), this proves $\{g_i\}$ is a $\tilde{\Theta}$-frame for X. The converse and the subset arguments follow straightforwardly by taking $\Theta \equiv \tilde{\Theta}$. If a larger solid BK-space $\Theta \supseteq \tilde{\Theta}$ exists where $\{g_i\}$ is a Θ-

frame with bounds A, B, then for $c \in \Theta$ and $f \in M_c$, solidity implies $|||c|||_\Theta \leq |||\{g_i(f)\}|||_\Theta \leq B\|f\|$. Taking the infimum yields $|||c|||_\Theta \leq B|||c|||_{\tilde{\Theta}}$. Since $A\|f\| \leq |||\{g_i(f)\}|||_\Theta \leq B|||\{g_i(f)\}|||_{\tilde{\Theta}}$, the lower bound naturally maps to the new space.

(c) If $\{g_i\}$ is a Banach frame for X with respect to Θ, let $V : \Theta \to X$ be the bounded reconstruction operator satisfying $V(\{g_i(f)\}) = f$. For every $c \in \tilde{\Theta}$, we have $\|Vc\| \leq \|V\|||c|||_\Theta \leq B\|V\|||c|||_{\tilde{\Theta}}$. This ensures V restricted to $\tilde{\Theta}$ remains bounded, proving $\{g_i\}$ is a Banach frame with respect to $\tilde{\Theta}$.

(d) To show $\tilde{\Theta}$ is a CB-space if and only if (A2) holds, we assume (A2) is valid. Fix $c \in \tilde{\Theta}$ and $\epsilon > 0$. By (A2), there exists $k \in \mathbb{N}$ such that $|||c(k)|||_{\tilde{\Theta}} < \epsilon$. This forces $|||c(n)|||_{\tilde{\Theta}} < \epsilon$ for all $n \geq k$, meaning the finite partial sum $\sum_{i=1}^{n} c_i e_i$ converges to c in the norm of $\tilde{\Theta}$ as $n \to \infty$. The canonical vectors therefore form a Schauder basis. □

9.3 Concrete Applications of the Construction

To demonstrate the robustness of this constructed space framework, we apply it to a sequence that heavily repeats elements of an orthonormal basis, making it extremely redundant and categorically failing the standard Bessel condition for ℓ^2.

Proposition 33 (Class of $\tilde{\Theta}$-frames)**.** *Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space, $\{e_i\}_{i=1}^{\infty}$ be an orthonormal basis for X, and $\Theta = \ell^2$. Let $\{g_i\}_{i=1}^{\infty} \in (X^*)^{\mathbb{N}}$ be defined by block repetition:*

$$\begin{aligned} g_i(f) &:= t_i\langle f, e_1\rangle, \quad i = 1, 2, \ldots, k_1, \\ g_i(f) &:= t_i\langle f, e_j\rangle, \quad i = k_{j-1}+1, k_{j-1}+2, \ldots, k_j, \quad j \in \mathbb{N}, j > 1, \end{aligned}$$

where $k_j \in \mathbb{N}$ defines the block boundaries, and $t_j \in \mathbb{R} \setminus \{0\}$ for all $j \in \mathbb{N}$. If $\tilde{\Theta}$ is defined by Equation (2), then $\tilde{\Theta}$ is a CB-space and $\{g_i\}_{i=1}^{\infty}$ is a Banach frame for X with respect to $\tilde{\Theta}$.

Proof. Let $c = \{c_i\}_{i=1}^{\infty} \in \tilde{\Theta}$ and $d = \{d_i\}_{i=1}^{\infty} \in \tilde{\Theta}$. We define normalized maximal block coefficients:

$$\tilde{c}_1 := \max\left(\frac{|c_1|}{|t_1|}, \frac{|c_2|}{|t_2|}, \ldots, \frac{|c_{k_1}|}{|t_{k_1}|}\right), \quad \tilde{d}_1 := \max\left(\frac{|d_1|}{|t_1|}, \frac{|d_2|}{|t_2|}, \ldots, \frac{|d_{k_1}|}{|t_{k_1}|}\right), \tag{12}$$

$$\tilde{c}_j := \max_{k_{j-1}<i\leq k_j} \frac{|c_i|}{|t_i|}, \quad \tilde{d}_j := \max_{k_{j-1}<i\leq k_j} \frac{|d_i|}{|t_i|}, \quad j > 1. \tag{13}$$

For any $f \in M_c$ and $h \in M_d$, the definition $M_{\underline{c}}$ guarantees $|c_i| \leq |g_i(f)|$ and $|d_i| \leq |g_i(h)|$. Consequently, $\tilde{c}_i \leq |\langle f, e_i\rangle|$ and $\tilde{d}_i \leq |\langle h, e_i\rangle|$ for all $i \in \mathbb{N}$.

To satisfy (A1), we construct an element $r_{f,h} \in M_{c+d}$ such that $\|r_{f,h}\| \leq \|f\| + \|h\|$. We define $r_{f,h} := m_1 e_1 + m_2 e_2 + m_3 e_3 + \ldots$, where the coefficients m_j are given by the block maximums of the sum: $m_j = \max_{k_{j-1} < i \leq k_j} \frac{|c_i + d_i|}{|t_i|}$ (with $k_0 = 0$). Since $m_i \leq \tilde{c}_i + \tilde{d}_i$, and both sequences $\{\tilde{c}_i\}$ and $\{\tilde{d}_i\}$ belong to ℓ^2 (bounded by the basis coefficients of f and h), $\{m_i\}_{i=1}^{\infty} \in \ell^2$, guaranteeing $r_{f,h} \in X$. It is straightforward to check that $|c_i + d_i| \leq t_i m_j = t_i \langle r_{f,h}, e_j\rangle = g_i(r_{f,h})$, confirming $r_{f,h} \in M_{c+d}$. Using the Minkowski inequality in ℓ^2, we find $\|r_{f,h}\| \leq \|\{\tilde{c}_i + \tilde{d}_i\}\|_{\ell^2} \leq \|\{\tilde{c}_i\}\|_{\ell^2} + \|\{\tilde{d}_i\}\|_{\ell^2} \leq \|f\| + \|h\|$, verifying (A1).

To verify (A3), consider $f \in X$ and take an arbitrary $\tilde{f} \in M_{\{g_i(f)\}}$. By definition, $|g_i(f)| \leq |g_i(\tilde{f})|$ implies $|\langle f, e_i\rangle| \leq |\langle \tilde{f}, e_i\rangle|$ for all $i \in \mathbb{N}$. Thus, $\|f\|^2 = \sum |\langle f, e_i\rangle|^2 \leq \sum |\langle \tilde{f}, e_i\rangle|^2 = \|\tilde{f}\|^2$, satisfying (A3) with lower bound $A = 1$.

To verify (A2), for $c \in \tilde{\Theta}$ and $\epsilon > 0$, choose $p \in \mathbb{N}$ such that the tail sum $\sum_{i=p}^{\infty} \tilde{c}_i^2 < \epsilon$. By defining a shifted sequence b and finding an element $h \in X$ mapping to it, we can verify that $h \in M_{c(k)}$ for $k = k_{p-1}$ and $\|h\|^2 < \epsilon$, satisfying (A2). Thus, $\tilde{\Theta}$ is a CB-space.

Finally, we construct the bounded left inverse V. We define dual elements:

$$f_i := \begin{cases} \frac{1}{t_i} e_1 & \text{if } i = 1 \\ \frac{1}{t_i} e_j & \text{if } i = k_{j-1} + 1 \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

. By analyzing the coordinate functionals $z_i \in \tilde{\Theta}$, we find $|||z_i|||_{\tilde{\Theta}} = 1/\|g_i\|_{X^*}$. Defining $Vz_i = f_i$ yields $\|Vz_i\| = \|f_i\| \leq 1/\|g_i\|_{X^*} = |||z_i|||_{\tilde{\Theta}}$, proving that V is bounded on the basis elements. Since $\tilde{\Theta}$ is a CB-space, we extend V by continuity to all of $\tilde{\Theta}$. For any $f \in X$, $V(\{g_i(f)\}) = \sum g_i(f) f_i = f$, completing the proof that $\{g_i\}$ acts as a Banach frame for X with respect to the constructed space $\tilde{\Theta}$. □

10 Fréchet Spaces through Projective Limits

Moving beyond the relatively comfortable confines of a single norm, we confront a harsh reality: a vast number of crucial spaces in functional analysis like the Schwartz space of rapidly decreasing functions or various spaces of

ultradistributions, simply cannot be captured by Banach topologies. Their geometric structure is far too nuanced to be reigned in by just one measure of magnitude. Instead, we must embrace locally convex topologies, relying on infinite families of seminorms to dictate convergence. To get our bearings and eventually set the stage for Fréchet frames, we first need to lock down exactly what we mean by projective topologies and limits. [86, 93, 60, 12, 19, 48, 75]

10.1 Projective Topologies

We have seen projective topologies before in Section 7.4, so we will not dwell on them too much. However, below we try to give an intuitive understanding of Fréchet spaces as a projective limit and why this contruction is especially useful in defining frames in Fréchet spaces.

Imagine a vector space E over a field K, sitting at the center of a web. Radiating outward from E are linear maps f_α, each connecting E to a different locally convex space $(E_\alpha, \mathcal{T}_\alpha)$, where the index α belongs to some arbitrary set A. How do we equip our central space E with a topology that respects all these connections simultaneously?

The most natural approach is to demand the bare minimum: we want every single mapping f_α to be continuous, but we don't want to add any superfluous open sets. Mathematically, this translates to taking the coarsest possible topology on E that maintains the continuity of the entire family $\{f_\alpha\}_{\alpha \in A}$. This uniquely determined structure $\mathcal{T}$ is known as the **projective topology** with respect to the family $\{(E_\alpha, \mathcal{T}_\alpha, f_\alpha) : \alpha \in A\}$.

To see what this topology actually looks like from the inside, consider the lattice of topologies on E. Our projective topology $\mathcal{T}$ acts as the upper bound of the inverse-image topologies $f_\alpha^{-1}(\mathcal{T}_\alpha)$. Practically speaking, if you want to build a neighborhood of some point $x \in E$, you look at its images $x_\alpha = f_\alpha(x)$ across a finite number of the target spaces. You take a neighborhood U_α around each x_α in E_α, pull them all back to E via f_α^{-1}, and intersect them. Because our maps are linear and the target topologies $\mathcal{T}_\alpha$ are locally convex, this intersection process inherently guarantees that $\mathcal{T}$ itself will be a translation-invariant locally convex topology, complete with a base of convex, absorbing 0-neighborhoods.

But a topology isn't particularly useful for analysis unless it can distinguish between distinct points. We need it to be Hausdorff. Interestingly enough, whether $\mathcal{T}$ achieves this separation property depends entirely on how well the maps f_α "see" the space E. We mentioned the conditions for

such a space to be Hausdorff in Section 7.4, but here we give a more complete proof for it.

Theorem 34 (Hausdorff Condition for Projective Topologies)**.** *The projective topology $\mathcal{T}$ on E generated by the family $\{(E_\alpha, \mathcal{T}_\alpha, f_\alpha)\}_{\alpha \in A}$ is a Hausdorff topology if and only if for every non-zero element $x \in E$, there exists at least one index $\alpha \in A$ and a 0-neighborhood $U_\alpha \subset E_\alpha$ such that the image $f_\alpha(x)$ completely misses U_α.*

Proof. Let's unravel the forward direction first. Suppose we already know $\mathcal{T}$ is Hausdorff, and we pick some $x \in E$ where $x \neq 0$. The Hausdorff property guarantees the existence of a 0-neighborhood U in E that stubbornly refuses to contain x. By the very definition of the projective topology's neighborhood base, this U must contain a finite intersection of pulled-back neighborhoods. That is, there is a finite subset $H \subset A$ and corresponding 0-neighborhoods $U_\alpha \subset E_\alpha$ such that $\bigcap_{\alpha \in H} f_\alpha^{-1}(U_\alpha) \subseteq U$. Since x isn't in U, it definitely cannot be in that intersection. Consequently, there has to be at least one specific index $\alpha \in H$ where $x \notin f_\alpha^{-1}(U_\alpha)$. Stripping away the inverse image, this is identical to saying $f_\alpha(x) \notin U_\alpha$.

Flipping the argument around, assume the condition holds. If $f_\alpha(x) \notin U_\alpha$, then naturally x cannot belong to the inverse image $f_\alpha^{-1}(U_\alpha)$. Because $f_\alpha^{-1}(U_\alpha)$ is an open neighborhood of 0 in the projective topology $\mathcal{T}$ that excludes x, the space must be Hausdorff. □

Before we move into limits, it is worth lingering for a moment on how external spaces interact with E. Suppose you have another topological space F and a map $u : F \to E$. When is u continuous? The projective topology's definition makes this almost trivially easy to check: u is continuous if and only if the composition $f_\alpha \circ u$ is continuous from F into $(E_\alpha, \mathcal{T}_\alpha)$ for every single index α. You only ever have to check continuity against the constituent mappings. However, you can find a complete proof for this in Section 7.4.

10.2 Constructing the Projective Limit

The real power of projective topologies emerges when we apply them to a structured, hierarchical system of spaces. Let us introduce a directed index set A, ordered by a relation $\leq$. For every $\alpha \in A$, we have a locally convex space E_α. Crucially, these spaces talk to each other. Whenever $\alpha \leq \beta$, there

exists a continuous linear transition map (sometimes called a bonding map) $u_{\alpha\beta} : E_\beta \to E_\alpha$.

Instead of just looking at an isolated space E, we look at the massive product space $\prod_{\alpha \in A} E_\alpha$. A typical element here is a sprawling thread $x = (x_\alpha)$, containing one piece from every single space in the family. We want to isolate the threads that are "coherent", those where the transition maps perfectly translate the higher-index components into the lower-index ones.

We define the **projective limit** of the family $\{E_\alpha\}$, denoted by $\lim_{\leftarrow} u_{\alpha\beta} E_\beta$, as the subspace of the product space $\prod_\alpha E_\alpha$ consisting of all elements $x = (x_\alpha)$ that satisfy the compatibility constraint:

$$x_\alpha = u_{\alpha\beta}(x_\beta) \quad \text{whenever } \alpha \leq \beta. \tag{15}$$

How do we topologize this limit? We simply borrow the projective topology discussed in the previous section. We take the canonical projection maps $p_\alpha : \prod E_\gamma \to E_\alpha$, restrict them to our limit subspace, and call these restrictions f_α. The topology on $\lim_{\leftarrow} u_{\alpha\beta} E_\beta$ is then precisely the projective topology with respect to these restricted projections f_α.

A rather beautiful quirk of this setup is that topological completeness, a non-negotiable requirement for any rigorous frame theory, survives the limiting process entirely intact.

Theorem 35 (Completeness of Limits)**.** *The projective limit of any family of complete locally convex spaces is itself a complete locally convex space.*

Proof. Let $E = \lim_{\leftarrow} u_{\alpha\beta} E_\beta$ be our projective limit, and let $F = \prod_{\alpha \in A} E_\alpha$ denote the overarching product space. It is a standard result in point-set topology that the arbitrary product of complete spaces remains complete. So, we already know F is complete.

To show that E is complete, it suffices to prove that E is a closed subspace of F. Let's examine the compatibility condition that defines E. For any specific pair of indices (α, β) with $\alpha \leq \beta$, we can define a subspace $V_{\alpha\beta} \subset F$ consisting of all threads x where $x_\alpha - u_{\alpha\beta}(x_\beta) = 0$.

Notice that the mapping $x \mapsto x_\alpha - u_{\alpha\beta}(x_\beta)$ is just a combination of continuous projections and continuous bonding maps, meaning the mapping itself is continuous from F into E_α. Assuming our constituent spaces E_α are Hausdorff (which they are), the single point $\{0\}$ is closed in E_α. Therefore, the inverse image of $\{0\}$ under this continuous mapping—which is exactly the subspace $V_{\alpha\beta}$—must be closed in F.

Our entire projective limit E is just the intersection of all these pairwise compatibility subspaces: $E = \bigcap_{\alpha \leq \beta} V_{\alpha\beta}$. Because the arbitrary intersection of closed sets is closed, E is a closed subspace of the complete space F. Consequently, E is undeniably complete. $\square$

10.3 Nested Banach Spaces and Fréchet Sequence Spaces

Now we arrive at the centerpiece of our topological stage: Fréchet spaces. Traditionally, a Fréchet space is defined quite abstractly as a complete, metrizable locally convex space. While mathematically elegant, this definition is practically opaque when you are trying to compute frame bounds.

However, because a Fréchet space is metrizable, its topology can be strictly generated by a *countable* fundamental system of seminorms. This countability changes everything, allowing us to represent the abstract Fréchet space as a highly structured, sequential projective limit of familiar Banach spaces. We have given the proof of this result already in Section 7.4, Theorem 21.

With Theorem 21 established, we possess the architectural blueprint needed to build frames. Instead of dealing with an abstract Fréchet space, we will operate strictly within families of localized Banach spaces.

We define our target Fréchet space X_F by taking a sequence of separable Banach spaces $\{X_s, \|\cdot\|_s\}_{s \in \mathbb{N}_0}$ that satisfy a rigid projective nesting. Specifically, the norms must be monotonically increasing:

$$\|\cdot\|_0 \leq \|\cdot\|_1 \leq \|\cdot\|_2 \leq \dots \tag{16}$$

This implies continuous embeddings $X_0 \supset X_1 \supset X_2 \dots$. We define our Fréchet space as the infinite intersection:

$$X_F := \bigcap_{s \in \mathbb{N}_0} X_s \tag{17}$$

and we impose a crucial density constraint: X_F must be dense in each individual constituent space X_s. This density is not just a pedantic topological requirement; as we saw in Section 9, it is the exact mechanism that allows localized frame bounds established on X_F to continuously extend out to the broader base spaces X_s.

Because frames inherently map vectors into coefficient sequences, we must mirror this nested Banach architecture on the sequence side. We define a

corresponding family of sequence spaces $\{\Theta_s, |||\cdot|||_s\}_{s\in\mathbb{N}_0}$, where each Θ_s is a Banach sequence space (specifically a BK-space or a CB-space). These must satisfy the identical nesting conditions:

$$|||\cdot|||_0 \leq |||\cdot|||_1 \leq |||\cdot|||_2 \leq \dots \tag{18}$$

The resulting **Fréchet sequence space** is defined analogously as $\Theta_F = \bigcap_{s\in\mathbb{N}_0} \Theta_s$. If every Θ_s is a CB-space (meaning the canonical vectors $\{e_i\}$ form a Schauder basis), then finite sequences automatically populate Θ_F, rendering it naturally dense in each Θ_s. In this environment, any sequence $c = \{c_i\} \in \Theta_F$ admits a unique canonical representation $c = \sum c_i e_i$ that converges simultaneously across every single localized norm $|||\cdot|||_s$.

Topological Invariants: Properties (DN) and (Ω)

While any sequence of nested Banach spaces forms a Fréchet space, the specific geometry of that space can vary wildly. To classify these spaces and ultimately determine whether a sequence of functionals can synthesize the entire space, functional analysts rely on deep structural invariants known as properties (DN) and (Ω).

Let E be a Fréchet space defined by a fundamental system of seminorms $(\|\cdot\|_k)_{k\in\mathbb{N}}$. The space E is said to possess **property (DN)** if the topology is "dominated by a norm." Rigorously, this means there exists a pivotal index $p \in \mathbb{N}$ such that for any other index $q \in \mathbb{N}$, you can find an index $n \in \mathbb{N}$ and a constant $C > 0$ satisfying the interpolation inequality:

$$\|x\|_q^2 \leq C\|x\|_p\|x\|_n \quad \text{for all } x \in E. \tag{19}$$

When this holds, the p-th seminorm $\|\cdot\|_p$ actually functions as a true, continuous norm on the entire space E.

Conversely, **property (Ω)** dictates the geometric behavior of the dual space. To define it, we first introduce the localized dual extended norms:

$$\|y\|_k^* := \sup\{|y(x)| : \|x\|_k \leq 1\} \quad \text{for } k \in \mathbb{N}. \tag{20}$$

The Fréchet space E possesses property (Ω) if, for every index $p \in \mathbb{N}$, there exists an index $q \in \mathbb{N}$ such that for any $n \in \mathbb{N}$, we can find a parameter $\theta \in (0, 1)$ and a constant $C > 0$ such that the dual norms interpolate as:

$$\|y\|_q^{*2} \leq C\|y\|_p^{*1-\theta}\|y\|_n^{*\theta} \quad \text{for all } y \in E^*. \tag{21}$$

Why do we care about these abstract inequalities? Because they characterize exactly when a Fréchet space is structurally isomorphic to a complemented subspace of s, the space of rapidly decreasing sequences. As it turns out, verifying that a frame analysis operator's range is a nuclear space strictly requires these two properties to hold simultaneously.

Köthe Sequence Spaces: The Prototypical Example

To move from the abstract to the concrete, we construct sequence spaces using infinite matrices. Let $A = (a_{i,k})_{i,k\in\mathbb{N}}$ be an infinite Köthe matrix filled with strictly positive real numbers, where the entries increase monotonically along the rows: $a_{i,k} \leq a_{i,k+1}$ for all $i, k \in \mathbb{N}$.

For any fixed parameter $p \in [1, \infty)$ and an index $k \in \mathbb{N}$, we construct the localized Köthe sequence space $\lambda^{p,k}(A)$ consisting of all complex sequences $x = (x_i)$ where the weighted ℓ^p-norm is finite:

$$|||x|||_{p,k} := \left(\sum_{i=1}^{\infty} |x_i a_{i,k}|^p \right)^{1/p} < \infty. \tag{22}$$

Each $\lambda^{p,k}(A)$ is a rock-solid, completely well-behaved Banach sequence space. Because the matrix entries $a_{i,k}$ increase with k, the norms naturally stack: $|||\cdot|||_{p,k} \leq |||\cdot|||_{p,k+1}$. This is the exact projective nesting constraint we established in this sub-section.

If we look at the specific case where $p = 2$ and the matrix entries grow polynomially, defined by $a_{i,k} = i^k$, the spaces $\lambda^{2,k}((i^k))$ act as weighted sequence Hilbert spaces. Taking the projective limit of this specific family as $k \to \infty$ yields s, the renowned nuclear Fréchet space of rapidly decreasing sequences. By substituting general BK-spaces for these Köthe spaces, the abstract theory of projective limits translates into a highly operational framework. We have now laid the necessary topological foundation. We possess infinite nested sequences of Banach spaces and their sequence space counterparts, ready to host the localized inequalities that define Fréchet frames in the next section.

11 Frames in Fréchet Spaces

We have finally arrived at the heart of the matter. Having spent the previous sections carefully constructing the localized sequences of Banach spaces and

establishing the topological validity of projective limits, we can now anchor the frame concept within the Fréchet environment. The challenge here is formidable. We are no longer dealing with a single, well-behaved norm that dictates convergence. Instead, we must wrangle an infinite, nested hierarchy of norms, ensuring that our frame inequalities do not just hold in isolation but communicate consistently across the entire projective spectrum.

To pull this off, the standard definitions of Banach frames must undergo a radical structural evolution. We cannot simply map a Fréchet space into a single sequence space; the topology would shatter. Rather, the topological straitjacket demands that the target sequence space itself must be a Fréchet space, woven together from an analogous projective limit of Banach sequence spaces. [77, 14, 18, 50, 59, 2]

11.1 Formulating the Fréchet Frame

Before we write down the definition, let us set the stage. We assume the existence of a sequence of separable Banach spaces $\{X_s, \|\cdot\|_s\}_{s\in\mathbb{N}_0}$ that are monotonically nested, meaning $X_0 \supset X_1 \supset X_2 \dots$ and the localized norms satisfy $\|\cdot\|_0 \leq \|\cdot\|_1 \leq \|\cdot\|_2 \dots$. Our primary domain is the Fréchet space $X_F := \bigcap_{s\in\mathbb{N}_0} X_s$, which we rigidly require to be dense in every single constituent space X_s. Parallel to this, we construct the target environment: a sequence of BK-spaces $\{\Theta_s, |||\cdot|||_s\}_{s\in\mathbb{N}_0}$ satisfying identical nesting constraints, yielding the Fréchet sequence space $\Theta_F := \bigcap_{s\in\mathbb{N}_0} \Theta_s$.

With the architecture in place, we can define what it actually means to capture a Fréchet space using a countable family of functionals.

Definition 36 (Pre-Fréchet Frame)**.** A sequence of continuous linear functionals $\{g_i\}_{i=1}^{\infty} \in (X_F^*)^{\mathbb{N}}$ is called a **pre-Fréchet frame** (or simply a **pre-F-frame**) for X_F with respect to Θ_F if, for every single integer index $s \in \mathbb{N}_0$, we can locate positive constants $0 < A_s \leq B_s < \infty$ such that two things happen:

1. The coefficient sequence unconditionally lands in the target space: $\{g_i(f)\}_{i=1}^{\infty} \in \Theta_F$ for all $f \in X_F$.

2. The frame inequalities hold locally across every tier of the projective limit: $A_s\|f\|_s \leq |||\{g_i(f)\}_{i=1}^{\infty}|||_s \leq B_s\|f\|_s$ for all $f \in X_F$.

If only the upper inequality holds alongside the containment condition, the sequence acts as an **F-Bessel sequence** for X_F with respect to Θ_F.

Notice what this definition forces upon the geometry of the space. The bounds A_s and B_s are allowed to drift as s increases, but the functional sequence $\{g_i\}$ itself is globally locked.

Of course, bounding the analysis operator is only half the battle. A frame isn't truly a frame unless you can run the process in reverse and synthesize the original vector back from its coefficient sequence. This brings us to the complete Fréchet frame.

Definition 37 (Fréchet Frame)**.** A pre-F-frame $\{g_i\}_{i=1}^{\infty}$ earns the title of a full **Fréchet frame (F-frame)** for X_F with respect to Θ_F if there exists an F-bounded, continuous linear reconstruction operator $V : \Theta_F \to X_F$ that perfectly recovers the signal, meaning $V(\{g_i(f)\}_{i=1}^{\infty}) = f$ for all $f \in X_F$.

To say the reconstruction operator V is "F-bounded" is a rather strong statement. It means that V does not just happen to be continuous on the Fréchet topology; rather, for every $s \in \mathbb{N}_0$, there exists a constant $K_s > 0$ such that $\|V(c)\|_s \leq K_s |||c|||_s$ for all coefficient sequences $c \in \Theta_F$. This strict, level-by-level bounded behavior is exactly what prevents the reconstruction process from tearing the projective limit apart.

11.2 Localizing the Frame Bounds

When you have a global property operating on a projective limit, an immediate question arises: do those properties trickle down cleanly to the localized base spaces? Because we smartly assumed that X_F is dense inside each X_s, the answer is yes. The density argument we deployed back in Section 9 works its magic here, allowing us to push the frame boundaries outward from X_F to the larger, localized Banach spaces.

Proposition 38 (Downward Extension of Bounds)**.** *Let $\{\Theta_s\}$ be a sequence of λ_s-BK-spaces satisfying our projective limit constraints. If $\{g_i\} \in (X_F^*)^{\mathbb{N}}$ acts as a pre-F-frame for X_F with respect to Θ_F possessing localized bounds A_s and B_s, then for any fixed $s \in \mathbb{N}_0$, the uniquely extended functionals $\{\tilde{g}_i^s\}_{i=1}^{\infty}$ function as a valid Θ_s-frame for the Banach space X_s, carrying the bounds A_s and $\lambda_s B_s$.*

Proof. Because X_F acts as a dense subspace within X_s, we can directly leverage the density extension mechanics proven earlier. Since $\{g_i\}$ is a pre-F-frame, we already know that for any $f \in X_F$, the frame inequalities $A_s\|f\|_s \leq |||\{g_i(f)\}|||_s \leq B_s\|f\|_s$ hold true. Because Θ_s is specifically a

λ_s-BK-space, the uniform boundedness of partial sums allows us to continuously push these inequalities from the dense set X_F into the outer edges of X_s. The unique continuous extension $\tilde{g}_i^s$ simply inherits the functional action of g_i, sealing the proof and verifying that the localized sequence constitutes a Θ_s-frame for X_s. □

This downward extension implies something rather beautiful about the analysis operator. The global operator $U : X_F \to \Theta_F$ defined by $f \mapsto \{g_i(f)\}$ natively splinters into a family of localized operators $U_s : X_s \to \Theta_s$, mapped by $f \mapsto \{\tilde{g}_i^s(f)\}$. Because these localized operators are bounded from below, the global range $R(U)$ is guaranteed to be a closed subspace within the Fréchet sequence space Θ_F.

11.3 The Duality Problem and DF-Frames

Standard Banach frame theory dictates that if you want to expand an element in the base space, you need a sequence residing in the dual space. Conversely, if you want to expand distributions in the dual space, you need elements from the base space. But transitioning to the dual of a Fréchet space (denoted X_F^*) introduces severe topological whiplash. The strong dual of a Fréchet space is not a Fréchet space; it is a completely different beast known as an (LB)-space, formed via the injective inductive limit of the dual Banach spaces X_s^*.

To tame this environment and permit valid signal reconstruction across both primal and dual spaces, we have to precisely define how dual frames operate within this inductive topology.

Definition 39 (DF-Bessel and DF-Frames)**.** Assume that the constituent sequence spaces Θ_s are all CB-spaces, which forces their duals Θ_s^* to act as BK-spaces. A sequence $\{f_i\}_{i=1}^{\infty} \in (X_F)^{\mathbb{N}}$ is classified as a **DF-Bessel sequence** for the dual space X_F^* with respect to Θ_F^* if it locally operates as a Θ_s^*-Bessel sequence for X_s^* across every level $s \in \mathbb{N}_0$.

If the sequence actually forms a Θ_s^*-frame for X_s^* at every level s, it is called a **pre-DF-frame**. Taking it one step further, if $\{f_i\}$ acts as a full Banach frame for X_s^* with respect to Θ_s^* across all $s \in \mathbb{N}_0$, it achieves the ultimate status of a **DF-frame** for X_F^* with respect to Θ_F^*.

11.4 Synthesis and Dual Expansions

We have reached the critical juncture. We know what a pre-F-frame looks like, and we understand the topological demands placed upon its reconstruction operator. Now, we must prove that when a pre-F-frame gets elevated to a full F-frame, it inherently spawns a dual sequence capable of completely expanding both functions and distributions.

This is the cornerstone result linking F-bounded projections to series expansions.

Theorem 40 (Existence of Dual Expansions)**.** *Let $\{g_i\}_{i=1}^{\infty}$ be a complete F-frame for X_F with respect to Θ_F. If the constituent sequence spaces Θ_s are all CB-spaces, then there exists a sequence $\{f_i\}_{i=1}^{\infty} \in (X_F)^{\mathbb{N}}$ acting as a DF-Bessel sequence for X_F^* with respect to Θ_F^* that provides the global expansions:*

$$f = \sum_{i=1}^{\infty} g_i(f) f_i, \quad \text{for all } f \in X_F, \text{ converging in } X_F, \text{ and} \tag{23}$$

$$g = \sum_{i=1}^{\infty} g(f_i) g_i, \quad \text{for all } g \in X_F^*, \text{ converging in } X_F^*. \tag{24}$$

Furthermore, the localized expansions $f = \sum \tilde{g}_i^s(f) f_i$ remain valid for all $f \in X_s$, and if the dual sequence spaces Θ_s^* are also CB-spaces, then $\{f_i\}_{i=1}^{\infty}$ is mathematically elevated to a pre-DF-frame.

Proof. Because $\{g_i\}$ is assumed to be an F-frame, we are guaranteed the existence of an F-bounded continuous linear reconstruction operator. By fundamental frame theory, this is topologically equivalent to stating that the range of the analysis operator, $R(U)$, is complemented in Θ_F via some F-bounded continuous projection P.

Fix an arbitrary index $s \in \mathbb{N}_0$. We know from the Proposition 38 that the extended sequence $\{\tilde{g}_i^s\}$ acts as a Θ_s-frame for X_s, meaning we have a bounded, localized analysis operator U_s. If we consider the projection P as an operator mapped from the dense subset $\Theta_F \subset \Theta_s$ into $R(U_s)$, we can uniquely and continuously extend it to a projection P_s defined on the entirety of Θ_s onto $R(U_s)$.

We can then define a localized inverse operator $V_s : \Theta_s \to X_s$ simply by setting $V_s := U_s^{-1} P_s$. Because V_s acts as a continuous extension of our original global inverse U^{-1}, we have essentially proven that the localized frame $\{\tilde{g}_i^s\}$ operates as a complete Banach frame for X_s with respect to Θ_s.

Now, we must extract the dual elements. We define our primal synthesis operator $V = U^{-1}P$, which is an F-bounded extension of U^{-1}. By applying this operator to the canonical unit vectors of the sequence space, we define $f_i := V(e_i)$ for all $i \in \mathbb{N}$. Let $f \in X_F$. Because Θ_F relies on CB-spaces, the sequence of coefficients $\{g_i(f)\}$ can be legitimately represented as $\sum_{i=1}^{n} g_i(f)e_i$. Using the stark continuity of V, we take the limit as $n \to \infty$:

$$V\left(\sum_{i=1}^{n} g_i(f)e_i\right) \to V(Uf) = f \quad \text{in } X_F. \tag{25}$$

Because V is linear, the left side is exactly $\sum_{i=1}^{n} g_i(f)V(e_i) = \sum_{i=1}^{n} g_i(f)f_i$, which proves the primal expansion $f = \sum_{i=1}^{\infty} g_i(f)f_i$. The dual expansion $g = \sum g(f_i)g_i$ naturally follows from the dual pairing properties across the (LB)-space. Since $V_s(e_i) = f_i$, the exact same logic applies across every X_s, yielding the localized expansions. Finally, invoking standard Banach frame duality, if both Θ_s and Θ_s^* are CB-spaces, the sequence $\{f_i\}$ natively inherits the lower frame bound across the dual spaces, proving it is a pre-DF-frame. □

We have established that possessing an F-frame guarantees the existence of a dual Bessel sequence that drives the series expansion. But in the world of functional analysis, one must always ask if the inverse holds true. If we stumble upon a sequence that allows us to expand every element in the Fréchet space, does that automatically force the generating sequence to be an F-frame?

The answer is a resounding yes, provided we have the baseline F-Bessel property.

Theorem 41 (Equivalence of F-Frames and Dual Expansions)**.** *Let Θ_s be CB-spaces, and let $\{g_i\}_{i=1}^{\infty}$ be an F-Bessel sequence for X_F with respect to Θ_F. There exists a sequence $\{f_i\} \in (X_F)^{\mathbb{N}}$ acting as a DF-Bessel sequence for X_F^* that satisfies the expansion $f = \sum g_i(f)f_i$ if and only if $\{g_i\}$ is a complete F-frame for X_F with respect to Θ_F.*

Proof. The forward implication that an F-frame yields the expansion was entirely handled by the previous result. The crux of the matter lies in proving the reverse direction: assume we possess a DF-Bessel sequence $\{f_i\}$ that successfully drives the expansion $f = \sum g_i(f)f_i$. We must prove that $\{g_i\}$ is an F-frame.

Fix an arbitrary level $s \in \mathbb{N}_0$. We already know that the localized analysis operator U_s is bounded. Let $f \in X_s$ and pick an arbitrary error threshold $\epsilon > 0$. Because the core Fréchet space X_F is relentlessly dense inside X_s, we can find some proxy element $h \in X_F$ such that $\|f - h\|_s < \epsilon$.

Applying our assumed expansion property to this proxy element h, there must exist an integer $N \in \mathbb{N}$ large enough such that $\|h - \sum_{i=1}^{n} g_i(h) f_i\|_s < \epsilon$ for all $n > N$. Let λ_s denote the canonical basis constant for the space Θ_s. Because the sequence $\{g_i(h - f)\}$ clearly belongs to Θ_s, we can bound the partial sum using the synthesis operator $T_s(d) = \sum d_i f_i$, which is guaranteed to be bounded from Θ_s into X_s because $\{f_i\}$ is a Θ_s^*-Bessel sequence.

For any $n > N$, we can brutally apply the triangle inequality:

$$\begin{aligned} \left\| f - \sum_{i=1}^{n} \tilde{g}_i^s(f) f_i \right\|_s &\leq \|f - h\|_s + \left\| h - \sum_{i=1}^{n} g_i(h) f_i \right\|_s + \left\| \sum_{i=1}^{n} \tilde{g}_i^s(h - f) f_i \right\|_s \\ &< \epsilon + \epsilon + \lambda_s \|T_s\| \|U_s\| \epsilon \\ &= \epsilon(2 + \lambda_s \|T_s\| \|U_s\|). \end{aligned}$$

Because ϵ was chosen arbitrarily, the error vanishes, and we are forced to conclude that $f = \sum_{i=1}^{\infty} \tilde{g}_i^s(f) f_i$ with strict convergence inside X_s. This localized convergence automatically verifies that $\{\tilde{g}_i^s\}$ acts as a Banach frame for X_s with respect to Θ_s.

To finalize the proof and upgrade the sequence to an F-frame, we must construct an F-bounded extension of the inverse operator U^{-1}. For any arbitrary coefficient sequence $d = \{d_i\} \in \Theta_F$, the synthesis series $\sum_{i=1}^{\infty} d_i f_i$ converges in X_s across every single level $s \in \mathbb{N}_0$, which strictly means it converges globally in X_F. The overarching synthesis operator $T(d) = \sum d_i f_i$ is thus perfectly defined from Θ_F into X_F and is s-bounded across all levels. Since the canonical vectors form a Schauder basis for Θ_F and T is continuous, applying it to the analysis sequence yields $T(Uf) = \sum g_i(f) T(e_i) = f$ for all $f \in X_F$. Because T acts as the F-bounded left inverse of U, the sequence $\{g_i\}$ is unequivocally an F-frame. $\square$

The theoretical implications of these two theorems cannot be overstated. By enforcing F-bounded continuity on the reconstruction operator (which is geometrically identical to demanding a complemented range for the analysis operator in the Fréchet sequence space), we entirely secure the existence of dual series expansions. We have successfully generalized the redundancy and

stability of frame decompositions from the safe, solitary norms of Banach spaces to the infinitely nested, highly restrictive topologies of Fréchet spaces.

12 Structural Generalizations

Up until now, we have forced our localized norms to march in perfect lockstep. In the standard, strict Fréchet frame setup outlined in the last section, the localized analysis operator is completely bounded above and below by the exact same index s. While mathematically elegant, this strict symmetry is something of a fantasy in hardcore functional analysis. Anyone who has spent time working with partial differential operators, or spaces of ultradifferentiable functions, knows that operators inherently shift regularity. You might be able to bound an expansion from above in one norm, but to get a stable lower bound, you often have to sacrifice a few derivatives, forcing you to look at a weaker norm.

This section is entirely about confronting that reality. We are going to intentionally break the symmetry of the frame inequalities. By relaxing the strict index matching, we arrive at the concepts of (X_1, Θ, X_2)-frames and general Fréchet frames. This structural generalization makes the theory wildly more robust, allowing it to capture highly asymmetrical operators that would categorically fail the strict conditions of standard Fréchet frames [78, 92, 15, 36, 36, 42].

12.1 (X_1, Θ, X_2)-Frames

Before we tackle the infinite nested structure of a Fréchet space, we need to understand how a frame behaves when it operates between two distinct, mismatched Banach spaces.

Suppose we have two Banach spaces $(X_1, \|\cdot\|_1)$ and $(X_2, \|\cdot\|_2)$ where one is continuously embedded inside the other. Specifically, let $X_2 \subseteq X_1$ and assume the norms satisfy the basic domination property $\|\cdot\|_1 \leq \|\cdot\|_2$. Now, introduce our target coefficient space, a BK-space $(\Theta, |||\cdot|||_\Theta)$.

Instead of bounding the frame strictly within a single space, we allow the lower bound to live in the larger space X_1, and the upper bound to pull from the smaller, more restrictive space X_2.

Definition 42 ((X_1, Θ, X_2)-frame)**.** A sequence of continuous linear functionals $\{g_i\}_{i=1}^{\infty} \in (X_2^*)^{\mathbb{N}}$ is classified as an (X_1, Θ, X_2)-frame if there exist

positive constants $0 < A \leq B < \infty$ such that the sequence of evaluations $\{g_i(f)\}_{i=1}^{\infty}$ unconditionally belongs to Θ for every $f \in X_2$, and the following shifted frame inequality holds:

$$A\|f\|_1 \leq |||\{g_i(f)\}_{i=1}^{\infty}|||_{\Theta} \leq B\|f\|_2, \quad \text{for all } f \in X_2. \tag{26}$$

If we were to collapse the spaces and set $X_1 = X_2 = X$, this definition would instantly snap back into a standard Θ-frame for X.

Just like we saw in Section 9, we rarely want to test these inequalities on every single vector in the space. Density arguments are our best friend here. But because we are dealing with two different topologies, we have to be careful about how the extension works.

Proposition 43 (Density Extension for Split Frames)**.** *Assume $(\Theta, |||\cdot|||_{\Theta})$ is a λ-BK-space. Let W be a dense subset of X_2. If the shifted frame inequalities $A\|f\|_1 \leq |||\{g_i(f)\}_{i=1}^{\infty}|||_{\Theta} \leq B\|f\|_2$ and the condition $\{g_i(f)\} \in \Theta$ are valid for all $f \in W$, then the sequence $\{g_i\}_{i=1}^{\infty}$ naturally extends to a full (X_1, Θ, X_2)-frame with slightly relaxed bounds A and λB.*

Proof. The proof relies heavily on the λ-BK-space structure. Because W is dense in X_2 under the stronger $\|\cdot\|_2$ norm, standard BK-space extension mechanics dictate that the upper inequality $|||\{g_i(f)\}|||_{\Theta} \leq \lambda B\|f\|_2$ will hold for all $f \in X_2$ and that the coefficient sequence remains inside Θ. We proved this exact mechanism back in Section 9.

The real trick is confirming the lower bound. Pick a stubborn element $f \in X_2 \setminus W$. Because W is dense in X_2, we can construct a sequence of approximations $\{f_n\} \in W^{\mathbb{N}}$ that converges to f in the $\|\cdot\|_2$ topology. Here is the critical observation: because $\|\cdot\|_1 \leq \|\cdot\|_2$, convergence in X_2 automatically guarantees convergence in X_1. So, $f_n \to f$ under the $\|\cdot\|_1$ norm as well.

Since the coordinate functionals are continuous, we know that

$$\lim_{n\to\infty} |||\{g_i(f_n)\}_{i=1}^{\infty}|||_{\Theta} = |||\{g_i(f)\}_{i=1}^{\infty}|||_{\Theta}.$$

We assumed the lower bound holds on W, so $A\|f_n\|_1 \leq |||\{g_i(f_n)\}_{i=1}^{\infty}|||_{\Theta}$ for every n. Pushing n to infinity, the limits perfectly preserve the inequality, yielding

$$A\|f\|_1 \leq |||\{g_i(f)\}_{i=1}^{\infty}|||_{\Theta}.$$

The extension is complete. □

12.2 General Pre-Fréchet Frames

Now let's escalate this concept to the full Fréchet topology. Remember our setup: we have a Fréchet space $X_F = \bigcap_{s\in\mathbb{N}_0} X_s$ dense in each Banach space X_s, alongside a Fréchet sequence space $\Theta_F = \bigcap_{s\in\mathbb{N}_0} \Theta_s$.

In a strict pre-F-frame, the norm indices had to match perfectly ($s = s$). To break this, we introduce two subsequences of indices, $\{s_k\}$ and $\{\tilde{s}_k\}$, both climbing to infinity. We demand that the lower bound index never exceeds the upper bound index, meaning $s_k \leq \tilde{s}_k$ for every integer k.

Definition 44 (General pre-Fréchet Frame)**.** A functional sequence $\{g_i\}_{i=1}^{\infty} \in (X_F^*)^{\mathbb{N}}$ is a **general pre-F-frame** for X_F with respect to Θ_F if there exist shifting subsequences $s_k \leq \tilde{s}_k$ and constants $0 < A_k \leq B_k < \infty$ such that the sequence lands in the target space $\{g_i(f)\}_{i=1}^{\infty} \in \Theta_F$, and the mismatched inequalities hold:

$$A_k\|f\|_{s_k} \leq |||\{g_i(f)\}_{i=1}^{\infty}|||_k \leq B_k\|f\|_{\tilde{s}_k}, \tag{27}$$

for all $f \in X_F,$ and all $k \in \mathbb{N}_0$.

This seemingly small tweak has massive implications. By letting the upper bound evaluate f using the much stronger norm $\|\cdot\|_{\tilde{s}_k}$, we are essentially allowing the frame elements g_i to behave like differential operators. They can "consume" regularity from f. As long as f is smooth enough (measured by the higher index $\tilde{s}_k$), the sequence of coefficients will still decay fast enough to sit inside Θ_k.

And just as before, because g_i is continuous and the Fréchet space is dense inside the base spaces, g_i possesses a unique continuous extension to $X_{\tilde{s}_k}$, which we denote as $g_i^{\tilde{s}_k}$. Consequently, at every localized step k, the extended sequence acts precisely as an $(X_{s_k}, \Theta_k, X_{\tilde{s}_k})$-frame.

12.3 The Reconstruction Problem and Generalized Expansions

Bounding the analysis operator $Uf = \{g_i(f)\}_{i=1}^{\infty}$ is nice, but frames are tools for synthesis. We need to be able to rebuild our signal f. In the strict setting, an F-bounded left inverse gave us expansions effortlessly. But now that our norms are decoupled, what exactly is required of our reconstruction operator $V : \Theta_F \to X_F$ to guarantee valid series expansions?

It turns out we need to aggressively control how V maps between the localized norms.

Theorem 45 (Conditions for Generalized Expansions)**.** *Let $\{g_i\}_{i=1}^{\infty}$ be a general pre-F-frame for X_F with respect to Θ_F, and assume the sequence spaces Θ_s are CB-spaces. The following statements are mathematically equivalent:*

(A1) *There exists a reconstruction operator $V : \Theta_F \to X_F$ satisfying $V(\{g_i(f)\}) = f$ for all $f \in X_F$, which is specifically bounded as follows: for every $k \in \mathbb{N}_0$, there is a constant $C_k > 0$ such that $\|Vd\|_{s_k} \leq C_k |||d|||_k$ for all $d \in \Theta_F$.*

(A2) *There exists a dual sequence $\{f_i\}_{i=1}^{\infty} \in (X_F)^{\mathbb{N}}$ that acts as a Θ_k^*-Bessel sequence for $X_{s_k}^*$ for every k, and guarantees the global primal expansion $f = \sum_{i=1}^{\infty} g_i(f) f_i$ unconditionally in X_F.*

(A3) *There exists a sequence $\{f_i\}_{i=1}^{\infty} \in (X_F)^{\mathbb{N}}$ that acts as a Θ_k^*-Bessel sequence for $X_{s_k}^*$, and drives the localized expansion $f = \sum_{i=1}^{\infty} g_i^{\tilde{s}_k}(f) f_i$ converging strictly in the weaker $\|\cdot\|_{s_k}$-norm for every $f \in X_{\tilde{s}_k}$.*

Proof. Let's unpack the mechanics of this, as the equivalence reveals the deep connection between operator boundedness and series convergence. We'll trace the path $(A1) \implies (A2), (A3)$ first.

Assume we have our bounded reconstruction operator V. We intuitively define our dual frame vectors by mapping the canonical basis of the sequence space: let $f_i = V(e_i)$ for all $i \in \mathbb{N}$. Because V is continuous on the Fréchet space, $V(\sum_{i=1}^{n} g_i(f) e_i)$ naturally converges to $V(Uf) = f$ as $n \to \infty$. This immediately hands us the global expansion in X_F.

Now, pick a specific index k. The boundedness condition on V means it can be extended to a bounded operator $V_k : \Theta_k \to X_{s_k}$. At the same time, our localized analysis operator U_k maps $X_{\tilde{s}_k} \to \Theta_k$. If we chain them together, the composition $V_k U_k$ maps $X_{\tilde{s}_k}$ into X_{s_k}. We know $V_k U_k f = f$ for elements inside the dense core X_F. Because the mapping is bounded and X_F is dense in $X_{\tilde{s}_k}$, this identity perfectly extends, meaning $V_k U_k f = f$ for all $f \in X_{\tilde{s}_k}$.

So if we take an element $f \in X_{\tilde{s}_k}$, we can write:

$$\sum_{i=1}^{n} g_i^{\tilde{s}_k}(f) f_i = V_k \left(\sum_{i=1}^{n} g_i^{\tilde{s}_k}(f) e_i \right).$$

As $n \to \infty$, the partial sum of canonical vectors converges in Θ_k, and since V_k is bounded, the entire expression converges to $V_k(U_k f) = f$ strictly within the $\|\cdot\|_{s_k}$ norm. This secures $(A3)$.

To see why $(A2) \implies (A1)$, imagine we are already holding the dual sequence $\{f_i\}$. Because it operates as a Θ_k^*-Bessel sequence for $X_{s_k}^*$, the synthesis operator $T_k(\{d_i\}) = \sum d_i f_i$ is perfectly well-defined and bounded from Θ_k into X_{s_k}. If we feed this operator a sequence from the Fréchet core Θ_F, the resulting series converges in X_{s_k} for every possible k. This means it converges in the Fréchet space X_F. We can then cleanly define $V(\{d_i\}) = \sum d_i f_i$. By standard operator properties, $\|Vd\|_{s_k} = \|T_k d\|_{s_k} \leq \|T_k\| \|\|d\|\|_k$. This matches the exact boundedness profile demanded by $(A1)$.

Finally, the jump from $(A3) \implies (A2)$ is almost trivial. The expansions in $(A3)$ hold for elements in the localized spaces $X_{\tilde{s}_k}$, which naturally completely cover the intersection space X_F. Thus, the local expansions instantly imply the global expansion required in $(A2)$. □

Extracting Subsequences

In Theorem 45, we placed a very specific, aggressive boundedness condition on V (bounding the s_k norm of the output against the k norm of the input). But what if V is just *continuous* from Θ_F to X_F? Do we lose all hope of localized expansions?

Not quite. Continuity on a Fréchet space simply means that a bound exists *somewhere* up the chain. It means we might skip a few indices, but we can still extract a valid subsequence of spaces where series expansions hold.

Theorem 46 (Subsequence Expansions via Continuity)**.** *Let $\{g_i\}_{i=1}^{\infty}$ be a general pre-F-frame. If there exists a continuous operator $V : \Theta_F \to X_F$ satisfying $V(\{g_i(f)\}) = f$, then we can extract increasing subsequences $\{w_j\}, \{r_j\}$, and $\{\tilde{w}_j\}$ such that the frame bounds hold locally as $\tilde{A}_j \|f\|_{w_j} \leq |||\{g_i(f)\}|||_{r_j} \leq \tilde{B}_j \|f\|_{\tilde{w}_j}$. Furthermore, there exists a sequence $\{f_i\}$ driving the localized expansion $f = \sum g_i^{\tilde{w}_j}(f) f_i$ strictly converging in the $\|\cdot\|_{w_j}$ norm for every $f \in X_{\tilde{w}_j}$.*

The proof of this is a fascinating exercise in index hunting. Because V is continuous, for every integer k, there has to exist some higher index p_k and a constant C_k such that $\|Vd\|_{s_k} \leq C_k |||d|||_{p_k}$. By defining a new pacing index $n_k = \max(k, p_k)$, we artificially force the bounds to line up. We then pull a strictly increasing subsequence n_{k_j} and apply the mechanics of Theorem 45

to this thinned-out hierarchy of spaces. We don't get expansions at every single level, but we get them on a co-final subsequence, which is entirely sufficient for synthesizing elements in the Fréchet core X_F.

12.4 General Fréchet Frames and the Complementedness Condition

With the synthesis mechanics mapped out, we can finally crown our generalized sequence with the title of a frame.

Definition 47 (General Fréchet Frame)**.** A sequence $\{g_i\}_{i=1}^{\infty} \in (X_F^*)^{\mathbb{N}}$ is declared a **general Fréchet frame (general F-frame)** for X_F with respect to Θ_F if it is a general pre-F-frame and there exists a continuous reconstruction operator $V : \Theta_F \to X_F$ perfectly inverting the analysis operator, $V(\{g_i(f)\}) = f$.

Just like in the strict case, the existence of this continuous reconstruction operator V is geometrically locked to the complementedness of the analysis operator's range.

Theorem 48. *Let $\{g_i\}_{i=1}^{\infty}$ be a general pre-F-frame. The existence of the continuous reconstruction operator $V : \Theta_F \to X_F$ is strictly equivalent to the existence of a continuous projection P from Θ_F onto the closed range $R(U)$.*

If the projection P exists, you simply define $V = U^{-1}P$. Because the general pre-F-frame condition guarantees that U^{-1} is continuous from $R(U)$ back to X_F, composing it with the continuous projection P yields a perfectly continuous reconstruction operator spanning the entirety of the sequence space. Conversely, if V is handed to you, the operator $P = UV$ acts as the necessary continuous projection onto $R(U)$.

The Harmonic Oscillator

To ensure this machinery isn't just theoretical vaporware, we apply it to a concrete, highly demanding environment: the Schwartz space $\mathcal{S}(\mathbb{R})$. We will construct a frame that systematically violates the strict F-frame definition but works flawlessly under the general F-frame umbrella.

Let ψ_j be the Hermite functions (the eigenfunctions of the harmonic oscillator), which form an orthonormal basis for $X_0 = L^2(\mathbb{R})$. We build our

Fréchet space X_F by defining the weighted Hilbert spaces X_s consisting of functions $\phi = \sum a_j \psi_j$ subject to the norm $\|\phi\|_s^2 = \sum |a_j|^2 j^{2s}$. The intersection of these spaces is precisely the Schwartz space $\mathcal{S}$. We mirror this on the sequence side with Θ_s, containing sequences where $\sum |d_j|^2 j^{2s} < \infty$, making Θ_F the space of rapidly decreasing sequences s.

Now, we define our frame elements. Let $r \in \mathbb{N}$ be a fixed penalty index. We define a highly irregular coefficient sequence b_j such that $|b_j| = 1$ for odd indices ($j = 1, 3, 5 \dots$) and $|b_j| = j^r$ for even indices ($j = 2, 4, 6 \dots$). We then define our functional sequence $g_j(\phi) = b_j \langle \phi, \psi_j \rangle$.

Does this sequence form a pre-F-frame? Let's check the norm bounds. Evaluating the coefficient sequence, we find:

$$||\phi||_s^2 \leq |||\{g_j(\phi)\}_{j=1}^{\infty}|||_s^2 = \sum_{j=1}^{\infty} |a_j b_j|^2 j^{2s} \leq ||\phi||_{s+r}^2. \tag{28}$$

The lower bound aligns perfectly with index s. But the upper bound completely blows past index s, forcefully requiring the heavier regularity of index $s + r$. Because of the j^r multiplier on the even terms, the sequence is structurally incapable of satisfying the strict frame inequalities $A_s \|\phi\|_{n_s} \leq |||\{g_j(\phi)\}|||_s \leq B_s \|\phi\|_{n_s}$ for any single matched index n_s.

If you apply the strict inequalities to an odd Hermite function, you get a contradiction on the upper bound. If you apply it to an even Hermite function, you get a contradiction on the lower bound. It categorically fails.

Yet, under Definition 44, this is a perfect general pre-F-frame. We simply set $s_k = k$ and $\tilde{s}_k = k + r$. Because the range of the analysis operator U naturally covers the entirety of Θ_F (meaning $R(U) = \Theta_F$), the reconstruction operator V is trivially just U^{-1}, which is continuous. Therefore, despite its wild, asymmetric bounds, the sequence $\{g_j\}$ functions mathematically as a pristine General Fréchet frame for the Schwartz space, entirely capable of driving unconditional series expansions.

This is the ultimate triumph of this section. By relaxing the topological symmetry and adopting shifted indices, we built a generalized framework that effortlessly captures differential-like multipliers and asymmetric decay profiles, bringing the full reconstructive power of frame theory to the untamed landscapes of non-Banach spaces.

13 Applications and Operator Properties

Up to this point, our journey through the architecture of Fréchet frames has dealt heavily with the abstract topological machinery required to make infinite families of localized norms behave nicely. We've wrestled with projective limits, sequences of Banach spaces, and the structural generalizations necessary to guarantee stable series expansions. But functional analysis is rarely just an abstract curiosity; these structures exist to solve real problems. In this final phase, we transition from the purely structural to the deeply operational. We will demonstrate how these frameworks directly govern operator invertibility, frame multipliers, and the synthesis of highly complex distribution spaces. [43, 79, 7, 8, 91, 40, 40, 37, 47]

13.1 Localized Frames and the Boundedness of Operators

When operating in signal processing or time-frequency analysis, one rarely deals with arbitrary frames. Practical frames usually exhibit some form of decay, their elements concentrate around specific points in time and frequency. To measure this, we don't look at the frame in isolation; rather, we measure its localization against a well-understood reference structure, typically a Riesz basis $G = (g_n)_{n=1}^{\infty}$ for the underlying Hilbert space $\mathcal{H}$.

We categorize this decay into three distinct regimes. A frame $E = (e_n)_{n=1}^{\infty}$ is considered:

- **Polynomially localized** with respect to G with decay $\gamma > 0$ if there exists a constant $C_\gamma > 0$ such that the cross-Gramian matrix entries obey the polynomial bound:

$$\max\{|\langle e_m, g_n\rangle|, |\langle e_m, \tilde{g}_n\rangle|\} \leq C_\gamma (1 + |m-n|)^{-\gamma}, \quad \text{for all } m, n \in \mathbb{N}. \tag{29}$$

- **Exponentially localized** if the decay is significantly faster, meaning there exist positive constants γ and C_γ such that:

$$\max\{|\langle e_m, g_n\rangle|, |\langle e_m, \tilde{g}_n\rangle|\} \leq C_\gamma e^{-\gamma|m-n|}, \quad \text{for all } m, n \in \mathbb{N}. \tag{30}$$

- **Sub-exponentially localized** (for a parameter $\beta \in (0, 1)$) if the decay sits between the polynomial and exponential regimes, satisfying:

$$\max\{|\langle e_m, g_n\rangle|, |\langle e_m, \tilde{g}_n\rangle|\} \leq C_\gamma e^{-\gamma|m-n|^\beta}, \quad \text{for all } m, n \in \mathbb{N}. \tag{31}$$

Why do we care about these specific decay rates? Because if a frame is localized with respect to a Riesz basis G, the frame operator S_E behaves beautifully not just on the Hilbert space $\mathcal{H}$, but on a family of associated Banach spaces. Given a sequence space Θ, we define the target Banach space $H^{\mu}_{\Theta G}$ via the reference basis G as the collection of elements whose coefficients are well-behaved:

$$H^{\mu}_{\Theta G} := \left\{ f \in \mathcal{H} : f = \sum_{n=1}^{\infty} c_n g_n \text{ with } (c_n)_{n=1}^{\infty} \in \Theta \right\}, \quad \|f\|_{H^{\mu}_{\Theta G}} := |||(c_n)_{n=1}^{\infty}|||_{\Theta}. \tag{32}$$

For a weight function μ satisfying the L^p condition (meaning $\mu(n) \geq 1$ and ℓ^p_{μ} embeds continuously into ℓ^2), the frame operator S_E preserves these spaces entirely.

Theorem 49 (Banach Invertibility and Dual Localization)**.** *Let $p \in [1, \infty)$ and suppose E is a frame for $\mathcal{H}$ that is appropriately localized with respect to a Riesz basis G (where the localization matches the decay of the weight μ—either polynomial, exponential, or sub-exponential). Then:*

1. *The frame operator S_E is an invertible, bounded bijection on $H^{\mu}_{\ell^p G}$.*
2. *The canonical dual frame $\tilde{E} = (\tilde{e}_n)_{n=1}^{\infty}$ inherits the exact same type of localization as the original frame E.*

Let's unpack the mechanics of this proof, because it reveals a profound trick in operator theory. We can define an infinite matrix $A_{m,n}$ tracking the decay profile of the frame (e.g., $A_{m,n} = (1+|m-n|)^{-(k+1+\epsilon)}$ for the polynomial case). When we apply the analysis operator U_E to an element $f \in H^{\mu}_{\ell^p G}$, we can bound the inner products strictly by exploiting the reference basis: $|\langle f, e_m \rangle| \leq C \sum_{n=1}^{\infty} A_{m,n} |\langle f, \tilde{g}_n \rangle|$. Standard matrix-weight lemmas confirm that this bounded matrix operation successfully maps the sequence into ℓ^p_{μ}. The synthesis operator T_E operates identically in reverse.

But how do we prove bijectivity? We construct an auxiliary operator $V := U_G S_E T_{\tilde{G}}$. Since this V acts entirely on the sequence space ℓ^2, we can invoke Jaffard's remarkable theorem on off-diagonal matrix decay. Jaffard's theorem guarantees that if a matrix is invertible on ℓ^2 and exhibits sufficient off-diagonal decay (which we established via $A_{m,n}$), its inverse matrix V^{-1} possesses the exact same decay profile. This forces V^{-1} to be a bounded bijection on ℓ^p_{μ}, which pulls backwards to prove that S_E is a bounded bijection

on $H^{\mu}_{\ell^p G}$. Because the canonical dual is formed by applying S_E^{-1}, the inherited matrix decay guarantees the dual elements perfectly retain the initial localization.

13.2 Invertibility on Schwartz and Ultradifferentiable Fréchet Spaces

The results on Banach spaces are impressive, but Phase 6 demands we push this into Fréchet topologies. If we treat the classic Hermite functions $(h_n)_{n=1}^{\infty}$ as our reference Riesz basis, the localized Banach spaces from the previous section naturally stack together via projective limits to form highly complex function spaces.

Specifically, if a frame decays polynomially with respect to the Hermite basis for every possible polynomial degree γ, it can traverse the entire projective hierarchy of the Schwartz space $\mathcal{S}(\mathbb{R})$.

Theorem 50 (Action on the Schwartz Space)**.** *If a frame $E = (e_n)_{n=1}^{\infty}$ for $L^2(\mathbb{R})$ is polynomially localized with respect to the Hermite basis for every $\gamma \in \mathbb{N}$, the frame operator S_E is a continuous bijection from $\mathcal{S}$ onto $\mathcal{S}$. Consequently, every element of the canonical dual frame resides inside the Schwartz space, $\tilde{e}_n \in \mathcal{S}$.*

But we don't have to stop at polynomial decay. By escalating to sub-exponential localization, the frame operator stabilizes over ultradifferentiable functions. Let Σ_α (for $\alpha > 1/2$) denote the Fréchet space of sub-exponentially decreasing functions, essentially representing Gevrey-class regularity.

Theorem 51 (Action on Ultradifferentiable Spaces)**.** *If a frame $E \subset \Sigma_\alpha$ for $L^2(\mathbb{R})$ is β-sub-exponentially localized (where $\beta = 1/(2\alpha)$) or exponentially localized with respect to the Hermite basis, then the frame operator S_E is a continuous bijection from Σ_α onto Σ_α. Furthermore, the canonical dual frame elements are ultradifferentiable, $\tilde{e}_n \in \Sigma_\alpha$.*

These theorems solidify a crucial narrative: if you design a frame with sufficient time-frequency decay, the brutal mathematics of Fréchet limits won't tear it apart. The canonical dual will natively mirror that smooth, rapidly decaying behavior.

13.3 Frame Multipliers

Up to this point, we have relied exclusively on the canonical dual frame $\tilde{E} = S_E^{-1}E$ to synthesize signals. While mathematically pristine because it minimizes the ℓ^2-norm of the expansion coefficients, the canonical dual is notoriously rigid. In certain applications, calculating it is computationally devastating, or it might lack specific traits (like compact support) that alternative duals possess. To break free from the canonical dual, we must introduce generalized operator modification via frame multipliers.

Definition 52 (Frame Multiplier)**.** Given two frames $\Phi = (\varphi_n)_{n=1}^{\infty}$ and $\Psi = (\psi_n)_{n=1}^{\infty}$ for $\mathcal{H}$, and a bounded scalar sequence $m = (m_n)_{n=1}^{\infty}$ acting as a symbol, the frame multiplier $M_{m,\Phi,\Psi} : \mathcal{H} \to \mathcal{H}$ is the bounded operator defined by:

$$M_{m,\Phi,\Psi} f = \sum_{n=1}^{\infty} m_n \langle f, \psi_n \rangle \varphi_n. \tag{33}$$

Multipliers allow us to filter, threshold, or amplify a signal directly through the frame coefficients. But the million-dollar question in multiplier theory is: if a frame multiplier is invertible, how do we represent its inverse? Does it ruin the frame structure?

A monumental theorem in this space dictates that the inverse of a multiplier is simply another multiplier, utilizing the reciprocal symbol $1/m$, but operating on specialized, non-canonical dual frames.

Theorem 53 (Multiplier Inversion)**.** *Let the symbol m be semi-normalized ($0 < \inf |m_n| \leq \sup |m_n| < \infty$). If the multiplier $M_{m,\Phi,\Psi}$ is invertible on $\mathcal{H}$, then there exist unique specialized dual frames $\Phi^\dagger$ and $\Psi^\dagger$ such that:*

$$M_{m,\Phi,\Psi}^{-1} = M_{1/m,\Psi^d,\Phi^\dagger}, \quad \text{for any arbitrary dual frame } \Psi^d \text{ of } \Psi, \tag{34}$$

$$M_{m,\Phi,\Psi}^{-1} = M_{1/m,\Psi^\dagger,\Phi^d}, \quad \text{for any arbitrary dual frame } \Phi^d \text{ of } \Phi. \tag{35}$$

These unique duals are explicitly given by $\Phi^\dagger = ((M_{m,\Phi,\Psi}^{-1})^*(m_n\psi_n))_{n=1}^{\infty}$ and $\Psi^\dagger = (M_{m,\Phi,\Psi}^{-1}(m_n\varphi_n))_{n=1}^{\infty}$.

This is a structural masterclass. By fixing one specialized dual ($\Phi^\dagger$), the formula allows the second dual to be completely arbitrary. But a deeper concern remains: if we build these specialized dual frames, do they destroy the highly sought-after localization properties we spent Section 13.1 and **??** establishing?

Remarkably, no. The localization is mathematically bulletproof against multiplier inversion.

Theorem 54 (Localization of Specialized Duals)**.** *Let the multiplier $M_{m,E,F}$ be invertible on $\mathcal{H}$. If the initial frames E and F share the exact same type of localization (polynomial, exponential, or sub-exponential), then the unique specialized dual frames $E^\dagger$ and $F^\dagger$ generated by the multiplier strictly preserve this exact type of localization (though the bounding constants determining the decay may shift).*

Because the localization is preserved, the entire Fréchet space infrastructure we built seamlessly accepts these multipliers. If E and F are polynomially localized with respect to the Hermite basis, then the frame multiplier $M_{m,E,F}$ acts as a continuous bijection on the Schwartz space $\mathcal{S}(\mathbb{R})$. If the frames are sub-exponentially localized, the multiplier forms a continuous bijection directly on the ultradifferentiable space Σ_α. The specialized duals generated by the multiplier remain trapped inside their respective Fréchet topologies, allowing us to completely bypass the canonical dual while retaining all stability and convergence guarantees.

13.4 Gelfand-Shilov Spaces and GS-Frames

To cap off the evolution of Fréchet frames, we confront one of the most structurally complex topological environments in functional analysis: Gelfand-Shilov spaces. These spaces, specifically the Roumieu version denoted as $S^{\{p!^s\}}(\mathbb{R}^d)$ for $s \geq 1/2$, sit at the extreme edge of distribution theory.

Classically, Gelfand-Shilov spaces are defined as *inductive* limits. But to apply the rigorous pre-F-frame and F-frame architectures we built, we need a projective limit structure. A stunning mathematical pivot allows us to represent the Roumieu Gelfand-Shilov space simultaneously as a projective limit of an uncountable number of Hilbert spaces.

We begin by defining $\mathcal{R}$ as the directed set of all strictly positive, monotonically increasing sequences $(r_i)_{i=1}^\infty$ that diverge to infinity. For each sequence $(r_i) \in \mathcal{R}$, we generate an associated subordinate weight function $N_{(r_i)}(\rho) = \sup_{p\in\mathbb{N}} \ln^+(\frac{\rho^p}{p!^s \prod_{i=1}^p r_i})$.

Using these weight functions, we can construct an uncountable family of weighted Sobolev-type Hilbert spaces $S_{p!^s,(r_i)}$, defined by the inner product

and induced norm:

$$|\phi|_{s,(r_i)} = \left(\sum_{\alpha \in \mathbb{N}_0^d} \int_{\mathbb{R}^d} \frac{e^{N_{(r_i)}(|t|)}}{\alpha!^s \prod_{j=1}^{|\alpha|} r_j} |D^\alpha \phi(t)|^2 dt \right)^{1/2} < \infty. \tag{36}$$

By taking the projective limit over the entire uncountably infinite directed set $\mathcal{R}$, we recover the Gelfand-Shilov space:

$$S^{\{p!^s\}}(\mathbb{R}^d) = \operatorname{proj} \lim_{(r_i) \in \mathcal{R}} S_{p!^s,(r_i)}. \tag{37}$$

Mirroring this construction on the sequence side, we define the constituent Hilbert sequence spaces $\Theta_{p!^s,(r_i)}$ containing sequences $(c_n)_{n=1}^\infty$ that satisfy the weighted decay condition $\sum_{n=1}^\infty |c_n|^2 e^{2N_{(r_i)}(n)} < \infty$. Taking the projective limit yields the target Gelfand-Shilov sequence space $\Theta_{\{p!^s\}} = \operatorname{proj} \lim_{(r_i) \in \mathcal{R}} \Theta_{p!^s,(r_i)}$.

With the environment successfully mapped into a projective hierarchy, we can finally define what it means to capture a Gelfand-Shilov space with a frame. Because the index set $\mathcal{R}$ is uncountable and highly volatile, the strict index matching of standard Fréchet frames fails entirely. We must use the shifted, asymmetric bound approach developed in Phase 5, giving rise to two distinct strengths of GS-frames.

Definition 55 (Weak GS-Frame (GFw-frame))**.** A sequence $(\varphi_n)_{n=1}^\infty \subset S^{\{p!^s\}}$ is a weak GS-frame for $S^{\{p!^s\}}(\mathbb{R}^d)$ with respect to $\Theta_{\{p!^s\}}$ if there exists a specific configuration of index sequences $(m_i^k), (r_i^k), (s_i^k) \in \mathcal{R}$ (where the sequences grow monotonically with k) such that the localized frame inequalities are satisfied:

$$A_k \|f\|_{(m_i^k)} \leq |||\{\varphi_n(f)\}_{n=1}^\infty|||_{(r_i^k)} \leq B_k \|f\|_{(s_i^k)}, \tag{38}$$

for all $k \in \mathbb{N}$, and $f \in S^{\{p!^s\}}$. Furthermore, there must exist a continuous synthesis operator $V : \Theta_{\{p!^s\}} \to S^{\{p!^s\}}$ capable of perfect reconstruction, $V((\varphi_n(f))_{n=1}^\infty) = f$.

The weak GS-frame guarantees that *some* path exists through the uncountable web of norms. But a stronger, much more resilient condition demands that the frame can adapt to *any* target sequence space norm thrown at it.

Definition 56 (Strong GS-Frame (GFs-frame))**.** A sequence $(\varphi_n)_{n=1}^{\infty}$ is a strong GS-frame if for **every** possible target weight sequence $(r_i^k) \in \mathcal{R}$, there exist corresponding lower and upper bounding weights $(m_i^k), (s_i^k) \in \mathcal{R}$ satisfying the exact same generalized frame inequalities and continuous reconstruction conditions detailed in the weak definition.

These final definitions represent the absolute summit of frame theory in locally convex spaces. By manipulating uncountably infinite projective limits, relaxing the strict symmetries of norm bounding, and utilizing continuous left inverses, the theories that began humbly in Hilbert space bases have been successfully morphed to synthesize elements in the most violently constrained spaces known to distribution theory.

14 Frames in p-Banach Spaces

Leaving behind the very structured and totally predictable borders of locally convex spaces makes a huge shift in our functional analysis approach completely necessary. Convexity getting stripped away means that building redundant and stable series expansions becomes a really delicate math wire-walk. Mapping this exact territory properly means looking strictly at how frame systems and atomic decompositions actually work mechanically. They are built specifically for p-Banach spaces. Here, the parameter p is stuck living in the restrictive interval $0 < p \leq 1$. Much of the charting done here comes from what Pahari, Kohli and Ghimre (2022) did. Other standard literature about atomic decompositions is also used [25, 74, 36, 20, 17, 64].

14.1 p-Norms

We really need to understand the warped geometry we are standing on before any kind of frame construction is attempted. It operates under rules that slightly twist standard Banach space axioms when you have a p-normed linear space X. Strictly non-negative is what the function $\|\cdot\|_p : X \to \mathbb{R}$ stays. It evaluates to zero if and only if the vector underneath it is literally zero itself. Scalar multiplication, however, does not scale in a linear way. Stretching a vector x by a scalar α instead yields $\|\alpha x\|_p = |\alpha|^p \|x\|_p$. The triangle inequality tightens down into a subadditive property similarly. It gets cleanly stated as $\|x + y\|_p \leq \|x\|_p + \|y\|_p$.

We officially call it a p-Banach space if X happens to be complete under this weird p-norm.

Calculating operator bounds looks pretty strange at first glance because of how scalar multiplication behaves non-linearly. Bounding a linear operator $T : X \to Y$ (where Y is assumed to be a q-Banach space) means you have to find a constant $M > 0$. This constant has to satisfy the fractional inequality $\|T(x)\|_q^{1/q} \leq M\|x\|_p^{1/p}$. Things simplify when the focus is narrowed to continuous linear functionals inside the conjugate space X^*. There being some M that ensures $|f(x)| \leq M\|x\|_p^{1/p}$ across all $x \in X$ is what makes a functional f bounded. Providing the norm for the dual space is what the supremum over the unit ball does then. This dual space still stays a standard Banach space itself.

14.2 Fundamental Sequences and Associated Spaces

Reconstructing elements from a space requires that your starting functional lineup can see all the non-zero vectors. A sequence $\{f_n\} \subset X^*$ gets declared as being *fundamental* over X if x being literally the zero vector is the only possible scenario where $f_n(x) = 0$ for every single integer n.

Power is granted incredibly by having a fundamental sequence in your pocket. The ability to artificially manufacture a sequence space mirroring your original p-Banach space perfectly is what it gives you.

Theorem 57 (Constructing the Associated Space)**.** *If $\{f_n\}$ acts fundamentally over a p-Banach space X, an associated p-Banach sequence space $X_d = \{\{f_n(x)\} : x \in X\}$ can always be constructed. It comes naturally endowed with the norm $\|\{f_n(x)\}\|_{X_d} = \|x\|_p$.*

Proof. An arbitrary $x \in X$ should be considered. Every functional naturally goes to zero if $x = 0$. The fundamental property rigidly makes sure that at least one functional gives an evaluation that is not zero if $x \neq 0$. Mapping every vector inside X to a very distinct sequence of scalars inside X_d is allowed by this. Verifying that this whole setup respects every foundational rule of a p-normed linear space is easy to do.

Proving completeness is the really critical hurdle. Let us imagine a Cauchy sequence of functional sequences existing, and we will call it $\{\{f_n(x_i)\}_n\}_i$. It lives inside X_d. Forcing the underlying sequence $\{x_i\}$ to act as a Cauchy sequence back inside the original space X is what the very way we defined our

new norm does immediately. Having no choice but to converge to some target element $x \in X$ is the fate of $\{x_i\}$ since X was assumed to be a complete p-Banach space. The sequence of sequences converging natively to $\{f_n(x)\}$ inside X_d happens consequently. This makes X_d complete in a way that cannot be denied. $\square$

The original space X being completely isomorphic to X_d via the simple isometric mapping $T(x) = \{f_n(x)\}$ is a direct and rather beautiful byproduct from this exact construction.

14.3 Atomic Decompositions and X_d-Frames

Formulating the core structure can happen now that the architectural foundation is laid down. Representing vectors is not the only thing a frame is about. Doing that representation while having redundancy and built-in stability is what it really is about. Building an atomic decomposition successfully for a p-Banach space X relative to its associated sequence space X_d requires you to have a paired system. This system is vectors $\{x_n\} \subset X$ and functionals $\{f_n\} \subset X^*$.

Earning the title of an atomic decomposition means this pair is strictly required to satisfy three non-negotiable pillars:

1. **Containment:** Safe landing in the target space by the coefficient sequence is needed. This means $\{f_n(x)\}_{n=1}^{\infty} \in X_d$ for every $x \in X$.

2. **Topological Bounds:** Disparate topologies must be bridged by existing positive constants A and B. Satisfying the localized inequality $A\|x\|_p \leq \|\{f_n(x)\}_n\|_{X_d} \leq B\|x\|_p$ is what they must do.

3. **Perfect Synthesis:** Putting the pieces back together to yield the unconditionally convergent series $x = \sum_{n=1}^{\infty} f_n(x)x_n$ is something you must be able to do.

Dropping that third requirement and abandoning the explicit reconstruction formula altogether downgrades the sequence $\{f_n\}$ structurally into an *X_d-frame*. This happens if you strictly demand only that the coefficients land inside X_d while the topological bounds are satisfied. Knowing X is isomorphic to a subspace of X_d mathematically giving a guarantee that an X_d-frame must exist is very interesting.

How intimately atomic decompositions get tied to the partial sum operators and their mechanical behavior is one of the most profound realizations in this non-locally convex setting. A dense subset A containing all finite linear combinations of our synthesis formula, $A = \{\sum_{i=1}^{n} f_i(x)x_i\}$, is what we should look at now.

Theorem 58 (Operator Equivalence)**.** *If the closure of A is the space X, the pair $(\{x_n\}, \{f_n\})$ acting as an atomic decomposition mathematically equates to the projection operators $P_n(x) = \sum_{i=1}^{n} f_i(x)x_i$ being uniformly bounded everywhere across the space.*

Proof. Unraveling the forward direction first is what we will do. A legitimate atomic decomposition is what we assume $(\{x_n\}, \{f_n\})$ to be. Every element x getting perfectly represented by the infinite sum $\sum f_n(x)x_n$ happens by definition. Geometrically, this is completely identical to saying the partial sums $P_n(x)$ converge straight to x as $n \to \infty$. The sequence of numbers $\|P_n x\|_p$ is inherently bounded for each individual $x \in X$ simply because it converges. Invoking the Banach-Steinhaus theorem (also known as the uniform boundedness principle) shows that pointwise boundedness across a complete space guarantees the operators themselves are uniformly bounded. This implies there exists some global constant K where $\|P_n\| \leq K$.

Going in the other direction is an exercise in topological density that is masterful. The projections being uniformly bounded by K is our assumption. The partial sums eventually just lock firmly onto the element itself once n gets large enough for any element already sitting inside the dense set A. Making $P_n(x) \to x$ a trivial conclusion is what this does.

But regarding an arbitrary element x floating in X, what happens? A tiny error margin $\epsilon > 0$ that is strictly positive should be picked. Finding a proxy element $y \in A$ sitting very close to x is possible because A is dense. Specifically, within a distance of $\epsilon/3$ is where it sits. Some threshold M where $\|P_n y - y\|_p < \epsilon/3$ for all $n \geq M$ exists since knowing $P_n(y)$ converges to y is something we already do.

By exploiting the subadditivity of the p-norm alongside our uniform bound K very aggressively, the triangle inequality can be wrangled. This wrangling forces $P_n(x)$ to converge to x across the entirety of X. Establishing the reconstruction formula $x = \sum f_i(x)x_i$ fully is what this accomplishes. The upper bound of the sequence norm getting checked reveals $\|\{f_n(x)\}\|_{X_d} \leq K\|x\|_p$. The remaining atomic decomposition criteria get satisfied by this. $\square$

The individual elements and functionals are trapped inside a surprisingly tight relationship if you secure an atomic decomposition that is heavily bounded by K. This is a quick structural consequence of the convergence. It satisfies $\|x_n\|_p \|f_n\|^p \leq 2K$.

14.4 Shrinking Decompositions and the Dual Space

Trying to look at how these representations interact with the conjugate space X^* makes things get noticeably more complicated. When mapped outwards, not all atomic decompositions behave equally well. Allowing us to effectively flip our perspective and expand elements sitting inside the dual space is what makes us label an atomic decomposition as *shrinking*.

Theorem 59 (Shrinking Duality)**.** *Operating as a valid atomic decomposition for the dual space X^* relative to an associated sequence space Z_d by the flipped pair $(\{f_n\}, \{\pi x_n\})$ happens if and only if an atomic decomposition $(\{x_n\}, \{f_n\})$ is shrinking. (Here, representing the natural embedding into the bidual X^{**} is what π does).*

Proof. Genuinely shrinking is what we suppose our initial decomposition does. Naturally obtaining $f(x) = \sum_{n=1}^{\infty} f(x_n) f_n(x)$ happens if we take any functional $f \in X^*$ and apply it directly to our standard reconstruction formula. Functionally identical to $\pi x_n(f)$ is what $f(x_n)$ is when adopting the natural embedding notation. Into the dual expansion $f(x) = \sum_{n=1}^{\infty} \pi x_n(f) f_n(x)$ is how the formula gracefully restructures itself thus.

Ensuring the series actually converges inside the topology of X^* is simply what we have to do to prove this forms a legitimate atomic decomposition for the dual space. The norm can be bounded if we isolate the tail end of the sum from index N to $M - 1$. Investigating how it operates on vectors trapped inside the unit ball does this: $\sup_{\|x\|_p \leq 1} |(f - S_N)(x)|$. Here, the truncated partial sum operator is represented by S_N. Vanishing as N grows to infinity is what the norm $\|f - S_N\|$ is mathematically forced to do because we explicitly assumed the decomposition is shrinking. The flipped pair gets securely established as a robust atomic decomposition for the conjugate space because this guarantees convergence. □

15 Conclusion

Pushing the boundaries of traditional frame theory systematically is what we have done throughout this dissertation. Into the vastly more abstract domains of general metric linear spaces, Fréchet spaces, and non-locally convex p-Banach spaces, the focus has entirely shifted. This leaves behind the geometrically forgiving and highly structured environments found in Hilbert and Banach spaces. Inner product structures, self-duality, orthogonality, and unconditional convergence are things that the classical formulation of frames relied on heavily, historically speaking. Breaking down frequently in general metric linear spaces is what these geometric luxuries do. Settings completely lacking local convexity and F-spaces show this especially. Standard functional analysis techniques get rendered largely inadequate because dual spaces have a tendency to become totally trivial in situations like these. Inner products are not strictly depended upon by the core reconstruction and stability properties of frames. This is the primary hypothesis validated throughout this work ultimately. Anchoring them firmly by the topological architecture and intrinsic metric of the ambient spaces themselves is completely possible instead.

Fréchet Spaces and Projective Limits

Serving as a cornerstone for this whole theoretical expansion is the reinterpreting of Fréchet spaces. We can seamlessly represent an abstract Fréchet space as a sequential projective limit of separable Banach spaces. This is strictly because it possesses a topology generated by a countable fundamental system of seminorms and is metrisable. Leaning on a solitary norm is avoided when formulating frame theory thanks to this architectural blueprint. Adapting to an infinite and nested hierarchy of norms is what the mathematics is forced to do instead. Consistently communicating across a projective spectrum is something these norms absolutely must do. Standard Banach sequence spaces naturally had to be generalised into CB-spaces and BK-spaces by us. Adequately supporting these complex structures required this. Coordinate continuity getting guaranteed alongside the uniform boundedness of partial sums is exactly what happens with these specialised spaces. Continuously extending frame bounds that were verified merely on a dense subset out to the entire space is illustrated by the research to be possible. Utilising these specific structures makes it happen. Tailoring custom sequence

spaces ($\tilde{\Theta}$) through constructive methodologies is what the study presents for times when sequences fail to natively act like Bessel sequences relative to standard spaces. Absolutely any sequence that is capable of yielding valid series expansions can be accommodated mathematically, and this proves it.

Fréchet Frames and Generalised Topologies

Establishing the concept of a **pre-Fréchet frame** gets helped tremendously by taking full advantage of projective limits. Mapping the coefficient sequence right into a target Fréchet sequence space is demanded by this framework. Fulfilling localised inequalities across every single tier of the projective limit simultaneously is also required. An F-bounded and continuous linear reconstruction operator perfectly recovering the initial signal is necessitated if one wants to upgrade this to a fully fledged **Fréchet frame (F-frame)**. Demanding that the range of the analysis operator remains fully complemented within the Fréchet sequence space is geometrically equivalent to this bounded reconstruction operator actually existing. Dual distributions as well as dual series expansions for primal functions have their existence completely guaranteed by enforcing F-bounded continuity therefore.

Recognising that strictly symmetric frame bounds end up being quite limiting inside advanced functional analysis is perhaps one of the most profound breakthroughs found in this research. Strict index matching between upper and lower bounds becomes virtually impossible. This happens because operators inherently shift regularity whenever applications involve ultradifferentiable functions or partial differential operators. Introducing **General Fréchet frames** alongside (X_1, Θ, X_2)**-frames** is how the study attempts to address this specific hurdle. Topological symmetry gets deliberately relaxed by these structures because they permit shifted indices. Behaving similarly to differential operators that "consume" regularity out of a stronger norm just to satisfy coefficient decay is what the frame elements can do in doing so. Highly irregular and totally asymmetric bounds are captured effortlessly by such a generalisation. Using Hermite functions, it also successfully frames complex topologies like the Schwartz space.

Operator Applications and Multipliers

When applied straight to operator theory is when the true operational power belonging to these frame structures gets fully realised. The frame operator

acts as an invertible and continuous bijection sitting on associated Fréchet and Banach spaces if a frame is locally bounded against a reference Riesz basis. This requires specific decay rates, which could be sub-exponential, exponential, or polynomial. Perfectly preserving ultradifferentiability, the exact smooth and rapidly decaying behaviour of the primal frame is naturally inherited by canonical dual frames as a direct consequence.

Into the realm of **frame multipliers**, the research pushes this idea even further. Utilising specialised dual frames to generate another multiplier simply out of the inverse of an invertible multiplier is proven by it. Against multiplier inversion, these specialised duals are virtually bulletproof, strikingly enough. Tearing the underlying Fréchet topology apart does not happen while they strictly preserve the original time-frequency localisation. Formulating strong and weak **GS-frames** makes use of uncountable projective limits, pushing the theory to its absolute summit. Synthesising distributions inside Roumieu Gelfand-Shilov spaces is something these are uniquely capable of doing.

Frame Systems in non-locally convex p-Banach Spaces

Analysing frame systems inside **p-Banach spaces** (for $0 < p \leq 1$) is how the dissertation finally tackles the complete absence of local convexity. Scalar multiplication scales totally non-linearly. Also, the triangle inequality twists itself into a strange subadditive property in these specific spaces. Artificially manufacturing associated p-Banach sequence spaces requires **atomic decompositions** to rely heavily on fundamental sequences made of continuous linear functionals. This happens within such restrictive geometries. The uniform boundedness of its partial sum projection operators mathematically equating to the very existence of an atomic decomposition is clearly established by the study. Demanding sequence containment, perfect synthesis, and strict topological bounds is what that decomposition does.

Bridging a rather critical gap found within functional analysis is what this work successfully does ultimately. Thriving far beyond the traditional confines of Hilbert spaces is entirely possible for the robust redundancy and stability of frames. Showing this is achieved by carefully intertwining operator continuity, asymmetric bounds, projective limits, and topological invariants. Mapping out the totally untamed landscapes of non-locally convex and non-Banach metric spaces is done comfortably in doing so.

16 Further Scope of Research

While this dissertation systematically extends frame theory from classical Hilbert and Banach spaces into the abstract domains of Fréchet spaces, and non-locally convex p-Banach spaces, several critical avenues remain open for future investigation. The foundational results presented here highlight clear patterns of what carries over and what breaks down when traditional geometric luxuries are removed, paving the way for targeted future studies.

Computational and Numerical Implementations

The current research deliberately restricts its scope to the theoretical, analytical, and topological properties of generalised frames. Consequently, a natural progression for future research is the development of numerical and computational implementations for these structures. Constructing practical algorithms to compute Fréchet frames, General Fréchet frames, and Gelfand-Shilov frames (GS-frames) would bridge the gap between abstract topological vector spaces and applied signal processing or non-harmonic Fourier analysis. A suitable approach to do this would be to use the representations based computability theory of Weihrauch et al. The nature of this theory which heavily used ideas of descriptive set theory and traditional analysis suit our current methods very well, and would possibly lead to some very interesting theory.

Overcoming Trivial Dual Spaces and Convergence Issues

Significant structural limitations persist in general metric linear spaces. Spaces lacking local convexity, such as L^p spaces for $0 < p < 1$, frequently suffer from trivial or insufficient dual spaces. The absence of non-trivial continuous linear functionals severely restricts the classical construction of dual frames and standard functional analysis techniques. Future research must investigate alternative reconstruction mechanisms that can operate independently of these linear functionals. Furthermore, the lack of guaranteed unconditional convergence in general F-spaces poses an ongoing challenge for stable series expansions. Developing new summation techniques or weakened convergence criteria tailored to these restrictive geometries remains a pressing open question.

References

[1] http://mathnet.preprints.org/EMIS/journals/BSANU/31/r2007_06.pdf. Accessed: 2026-5-3.

[2] TI Abanina. On the problem of a frame concept. *Nauchnoe obozrenie*, 9:101–104, 2013.

[3] Fernando Albiac and Nigel J Kalton. *Topics in Banach space theory*. Springer, 2006.

[4] Günter Albinus. über lokal radialbeschränkte oder lokal pseudokonvexe metrisierbare Räume. *Math. Nachr.*, 45:363–372, 1970.

[5] Akram Aldroubi, Qiyu Sun, and Wai-Shing Tang. p-frames and shift invariant subspaces of L^p. *J. Fourier Anal. Appl.*, 7(1):1–21, 2001.

[6] Tosio Aoki. Locally bounded linear topological spaces. *Proc. Imp. Acad. Tokyo*, 18:588–594, 1942.

[7] Peter Balazs. Basic definition and properties of bessel multipliers. *Journal of Mathematical Analysis and Applications*, 325(1):571–585, 2007.

[8] Peter Balazs and Diana T Stoeva. Representation of the inverse of a frame multiplier. *Journal of Mathematical Analysis and Applications*, 422(2):981–994, 2015.

[9] Stefan Banach. Sur les fonctionnelles linéaires ii. *Studia Mathematica*, 1(1):223–239, 1929.

[10] Stefan Banach. Théorie des opérations linéaires. 1932.

[11] NK Bari. Sur les bases dans l'espace de hilbert. In *Dokl. Akad. Nauk SSSR*, volume 54, pages 379–382, 1946.

[12] Klaus-D Bierstedt, Reinhold Meise, and William H Summers. A projective description of weighted inductive limits. *Transactions of the American Mathematical Society*, 272(1):107–160, 1982.

[13] Garrett Birkhoff. A note on topological groups. *Compositio mathematica*, 3:427–430, 1936.

[14] José Bonet, Carmen Fernández, Antonio Galbis, and Juan Miguel Ribera. Frames and representing systems in fréchet spaces and their duals. 2017.

[15] Lasse Borup, Rémi Gribonval, and Morten Nielsen. Tight wavelet frames in lebesgue and sobolev spaces. *Journal of function spaces*, 2(3):227–252, 2004.

[16] Nicolas Bourbaki. *Topological vector spaces: Chapters 1–5*. Springer Science & Business Media, 2013.

[17] Daniel Carando and Silvia Lassalle. Duality, reflexivity and atomic decompositions in banach spaces. *Studia Math*, 191(1):67–80, 2009.

[18] Daniel Carando, Silvia Lassalle, and Pablo Schmidberg. The reconstruction formula for banach frames and duality. *Journal of Approximation Theory*, 163(5):640–651, 2011.

[19] P Pérez Carreras and José Bonet. *Barrelled locally convex spaces*, volume 131. Elsevier, 1987.

[20] Pete Casazza, Ole Christensen, and Diana T. Stoeva. Frame expansions in separable Banach spaces. *J. Math. Anal. Appl.*, 307(2):710–723, 2005.

[21] Peter G Casazza, Deguang Han, and David R Larson. Frames for banach spaces. *Contemporary Mathematics*, 247:149–182, 1999.

[22] Ole Christensen. *An introduction to frames and Riesz bases*. Applied and Numerical Harmonic Analysis. Birkhäuser Boston, Inc., Boston, MA, 2003.

[23] Ole Christensen and Christopher Heil. Perturbations of Banach frames and atomic decompositions. *Math. Nachr.*, 185:33–47, 1997.

[24] Ole Christensen and Diana T. Stoeva. p-frames in separable Banach spaces. volume 18, pages 117–126. 2003. Frames.

[25] Ronald R. Coifman and Guido Weiss. Extensions of Hardy spaces and their use in analysis. *Bull. Amer. Math. Soc.*, 83(4):569–645, 1977.

[26] Elena Cordero and Karlheinz Gröchenig. Localization of frames. II. *Appl. Comput. Harmon. Anal.*, 17(1):29–47, 2004.

[27] Ingrid Daubechies. The wavelet transform, time-frequency localization and signal analysis. *IEEE Transactions on Information Theory*, 36(5):961–1005, 1990.

[28] Ingrid Daubechies. *Ten lectures on wavelets*. SIAM, 1992.

[29] Ingrid Daubechies and Alex Grossmann. Frames in the bargmann space of entire functions. *Communications on Pure and Applied Mathematics*, 41(2):151–164, 1988.

[30] Ingrid Daubechies, Alex Grossmann, and Yves Meyer. Painless nonorthogonal expansions. *Journal of Mathematical Physics*, 27(5):1271–1283, 1986.

[31] Mahlon M. Day. The spaces L^p with $0 < p < 1$. *Bull. Amer. Math. Soc.*, 46:816–823, 1940.

[32] Richard J Duffin and Albert C Schaeffer. Power series with bounded coefficients. *American journal of mathematics*, 67(1):141–154, 1945.

[33] Richard J Duffin and Albert C Schaeffer. A class of nonharmonic fourier series. *Transactions of the American Mathematical Society*, 72(2):341–366, 1952.

[34] P. L. Duren, B. W. Romberg, and A. L. Shields. Linear functionals on H^p spaces with $0 < p < 1$. *J. Reine Angew. Math.*, 238:32–60, 1969.

[35] Hans G. Feichtinger and Karlheinz Gröchenig. A unified approach to atomic decompositions via integrable group representations. In *Function spaces and applications (Lund, 1986)*, volume 1302 of *Lecture Notes in Math.*, pages 52–73. Springer, Berlin, 1988.

[36] Hans G Feichtinger and Karlheinz H Gröchenig. Banach spaces related to integrable group representations and their atomic decompositions, i. *Journal of Functional analysis*, 86(2):307–340, 1989.

[37] Hans G Feichtinger and Krzysztof Nowak. A first survey of gabor multipliers. *Advances in Gabor analysis*, pages 99–128, 2003.

[38] Maurice Fréchet. Sur quelques points du calcul fonctionnel. *Rendiconti del Circolo Matematico di Palermo (1884-1940)*, 22(1):1–72, 1906.

[39] Maurice Fréchet. Les espaces abstraits topologiquement affines. *Acta Math*, 47:25–52, 1926.

[40] Izrail Moiseevich Gelfand, Georgiĭ Evgenevich Shilov, Morris D Friedman, Amiel Feinstein, and Christian P Peltzer. Spaces of fundamental and generalized functions. *(No Title)*, 2016.

[41] Karlheinz Gröchenig. Describing functions: atomic decompositions versus frames. *Monatsh. Math.*, 112(1):1–42, 1991.

[42] Karlheinz Gröchenig. Localized frames are finite unions of riesz sequences. *Advances in computational mathematics*, 18(2):149–157, 2003.

[43] Karlheinz Gröchenig. Localization of frames, banach frames, and the invertibility of the frame operator. *Journal of Fourier Analysis and Applications*, 10(2):105–132, 2004.

[44] Alexandre Grothendieck. Produits tensoriels topologiques et espaces nucléaires. *Mem. Amer. Math. Soc.*, 16:Chapter 1: 196 pp.; Chapter 2: 140, 1955.

[45] Hans Hahn. Über lineare gleichungssysteme in linearen räumen. 1927.

[46] Christopher Heil. *A basis theory primer: expanded edition*. Springer Science & Business Media, 2011.

[47] S. Jaffard. Propriétés des matrices "bien localisées" près de leur diagonale et quelques applications. *Ann. Inst. H. Poincaré C Anal. Non Linéaire*, 7(5):461–476, 1990.

[48] Hans Jarchow. *Locally convex spaces*. Springer Science & Business Media, 2012.

[49] W. B. Johnson and J. Lindenstrauss, editors. *Handbook of the geometry of Banach spaces. Vol. 2*. North-Holland, Amsterdam, 2003.

[50] VM Kadets. Representing and absolutely representing systems. *Studia Mathematica*, 102:217–223, 1992.

[51] Shizuo Kakutani. über die Metrisation der topologischen Gruppen. *Proc. Imp. Acad. Tokyo*, 12(4):82–84, 1936.

[52] N. J. Kalton. Quotients of F-spaces. *Glasgow Math. J.*, 19(2):103–108, 1978.

[53] N. J. Kalton, N. T. Peck, and James W. Roberts. *An F-space sampler*, volume 89 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 1984.

[54] S. K. Kaushik and S. K. Sharma. Atomic decompositions satisfying certain properties. *J. Adv. Res. Pure Math.*, 3(2):40–48, 2011.

[55] S. K. Kaushik and S. K. Sharma. On a generalization of atomic decompositions. *Albanian J. Math.*, 5(1):21–29, 2011.

[56] S. K. Kaushik and S. K. Sharma. On unconditional atomic decompositions in Banach spaces. *J. Appl. Funct. Anal.*, 6(4):343–355, 2011.

[57] VL Klee. Invariant metrics in groups (solution of a problem of banach). *Proceedings of the American Mathematical Society*, 3(3):484–487, 1952.

[58] A Kolomogroff. Zur normierbarkeit eines topologichen linearem raumes. *Studia Mathematica*, 5:29–33, 1934.

[59] Yu F Korobeinik. Representing systems. *Russian Mathematical Surveys*, 36(1):75–137, 1981.

[60] Yu F Korobeinik. Inductive and projective topologies. sufficient sets and representing systems. *Mathematics of the USSR-Izvestiya*, 28(3):529–554, 1987.

[61] Gottfried Köthe. Topological vector spaces. In *Topological Vector Spaces I*, pages 123–201. Springer, 1969.

[62] M. Krein and D. Milman. On extreme points of regular convex sets. *Studia Math.*, 9:133–138, 1940.

[63] Max Landsberg. Lineare topologische Räume, die nicht lokalkonvex sind. *Math. Z.*, 65:104–112, 1956.

[64] Rui Liu. On shrinking and boundedly complete Schauder frames of Banach spaces. *J. Math. Anal. Appl.*, 365(1):385–398, 2010.

[65] Stanisław Mazur and Ludwik Sternbach. Über die borelschen typen von linearen mengen. *Studia Mathematica*, 4(1):48–53, 1933.

[66] Reinhold Meise and Dietmar Vogt. *Introduction to functional analysis*. Clarendon press, 1997.

[67] Hermann Minkowski. Allgemeine lehrsätze über die konvexen polyeder. In *Ausgewählte Arbeiten zur Zahlentheorie und zur Geometrie: Mit D. Hilberts Gedächtnisrede auf H. Minkowski, Göttingen 1909*, pages 121–139. Springer, 1897.

[68] J. Musielak. *Modular Spaces*. Wyd. UAM, Poznań, 1978.

[69] J. Musielak and W. Orlicz. On modular spaces. *Studia Math.*, 18:49–65, 1959.

[70] J. Musielak and W. Orlicz. Some remarks on modular spaces. *Bull. Acad. Polon. Sci. Sér. Sci. Math. Astr. Phys.*, 7:661–668 (unbound insert), 1959.

[71] Hidegorô Nakano. *Modulared Semi-Ordered Linear Spaces*. Maruzen Co. Ltd., Tokyo, 1950.

[72] Władysław Orlicz. Über unbedingte konvergenz in funktionenraümen (ii). *Studia Mathematica*, 4(1):41–47, 1933.

[73] Władysław Orlicz. Über folgen linearer operationen, die von einem parameter abhängen. *Studia Mathematica*, 5(1):160–170, 1934.

[74] NP Pahari, Teena Kohli, and JL Ghimire. Frame systems in non-locally convex banach spaces. *Jordan Journal of Mathematics and Statistics*, 15(2):231–242, 2022.

[75] S Pilipović, Diana Stoeva, and N Teofanov. Frames for fréchet spaces. *Bulletin (Académie serbe des sciences et des arts. Classe des sciences mathématiques et naturelles. Sciences mathématiques)*, pages 69–84, 2007.

[76] Stevan Pilipović and Diana T Stoeva. On sequence spaces for fr\'echet frames. *arXiv preprint arXiv:0811.3182*, 2008.

[77] Stevan Pilipović and Diana T Stoeva. Series expansions in fréchet spaces and their duals, construction of fréchet frames. *Journal of Approximation Theory*, 163(11):1729–1747, 2011.

[78] Stevan Pilipović and Diana T Stoeva. Fréchet frames, general definition and expansions. *Analysis and Applications*, 12(02):195–208, 2014.

[79] Stevan Pilipović and Diana T Stoeva. Localization of fréchet frames and expansion of generalized functions. *Bulletin of the Malaysian Mathematical Sciences Society*, 44(5):2919–2941, 2021.

[80] Frédéric Riesz. *Sur les systèmes orthogonaux de fonctions (et l'équation de Fredholm)*. 1907.

[81] Alex P Robertson and Wendy Robertson. *Topological vector spaces*, volume 53. CUP Archive, 1980.

[82] S. Rolewicz. On a certain class of linear metric spaces. *Bull. Acad. Polon. Sci. Cl. III.*, 5:471–473, XL, 1957.

[83] S. Rolewicz. Some remarks on the spaces $N(L)$ and $N(l)$. *Studia Math.*, 18:1–9, 1959.

[84] S. Rolewicz and C. Ryll-Nardzewski. On unconditional convergence in linear metric spaces. *Colloq. Math.*, 17:327–331, 1967.

[85] Stefan Rolewicz. *Metric linear spaces*, volume 20 of *Mathematics and its Applications (East European Series)*. D. Reidel Publishing Co., Dordrecht; PWN—Polish Scientific Publishers, Warsaw, second edition, 1985.

[86] H. H. Schaefer and M. P. Wolff. *Topological vector spaces*, volume 3 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 1999.

[87] Laurent Schwartz. Théorie des distributions à valeurs vectorielles. i. In *Annales de l'institut Fourier*, volume 7, pages 1–141, 1957.

[88] J. H. Shapiro. Examples of proper, closed, weakly dense subspaces in non-locally convex F-spaces. *Israel J. Math.*, 7:369–380, 1969.

[89] S. Simons. Boundedness in linear topological spaces. *Trans. Amer. Math. Soc.*, 113:169–180, 1964.

[90] Ivan Singer. *Bases in Banach spaces II*. Springer, 1981.

[91] Diana Stoeva and STEVAN PILIPOVIĆ. Fréchet and gf frames and invertibility of frame multipliers on banach and fréchet frames. *Bulletin (Académie serbe des sciences et des arts. Classe des sciences mathématiques et naturelles. Sciences mathématiques)*, (49):5–18, 2024.

[92] Diana T Stoeva. Xd-frames in banach spaces and their duals. *Int. J. Pure Appl. Math*, 52(1):1–14, 2009.

[93] BA Taylor. Some locally convex spaces. *Entire Functions and Related Parts of Analysis: Proceedings*, 11:431, 1968.

[94] Philippe Turpin. Sur une classe d'algèbres topologiques. *C. R. Acad. Sci. Paris Sér. A-B*, 263:A436–A439, 1966.

[95] Philippe Turpin. Convexités dans les espaces vectoriels topologiques généraux. *Dissertationes Math. (Rozprawy Mat.)*, 131:221, 1976.

[96] Frederick A. Valentine. *Convex sets*. McGraw-Hill Series in Higher Mathematics. McGraw-Hill Book Co., New York-Toronto-London, 1964.

[97] Robert M Young. *An introduction to non-Harmonic fourier series, revised edition, 93*. Elsevier, 2001.

[98] W. Żelazko. Metric generalizations of Banach algebras. *Rozprawy Mat.*, 47:70, 1965.